The Möbius function at $N - p$

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Abstract

For even $\mathcal{N}$ tending to infinity, we deduce cancellation of $\mu(\mathcal{N} - p)$ over primes $p < \mathcal{N}$, and equal densities of its two nonzero signs, from a zero-free region implied by the 7/8 half-plane of OpenAI and the arguments of Lü, Wang and Wu as written. We give the changes to their determinant method for this varying shift. For each fixed C, Iwaniec's 1974 zero-free region and Burgess's bound supply the analytic input when $\mathrm{rad}(\mathcal{N}) \le (\log \mathcal{N})^C$. Assuming the additional arguments of Lü, Wang and Wu as written, we also obtain cancellation uniformly for $0 < |h| \le (\log x)^C$ in $\mu(p + h)$ on positive arguments. The corresponding Liouville statements have sign densities $1/2 + o(1)$.

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1 Introduction

All logarithms are natural. The Möbius function is denoted by μ, and $\lambda(n) = (-1)^{\Omega(n)}$ is the Liouville function on positive integers. Here Ω counts prime factors with multiplicity. Write $\mathrm{rad}(n) = \prod_{p \mid n} p$ and let $\pi(x)$ count the primes at most x. Put

$$A_{\mathcal{N}} = \prod_{p \nmid \mathcal{N}} \left(1 - \frac{1}{p(p-1)}\right), \qquad A_h = \prod_{p \nmid h} \left(1 - \frac{1}{p(p-1)}\right). \tag{1.1}$$

These products are at least Artin's constant $0.3739\ldots$.

We specify the inputs to the statements. The assertion H2′ is that the arguments of LWW Sections 4.2–4.7, including the proofs of Propositions 4.3–4.5 and Proposition 4.2, and Appendix A, (A.2)–(A.4), are correct as written [6, pp. 17–39, 48–50]. The assertion H1 is

$$L(s, \chi) \neq 0 \quad (\Re s > 7/8) \quad \text{for every Dirichlet character } \chi, \tag{1.2}$$

with the principal pole at $s = 1$ allowed. This is the Dirichlet conclusion of [10, Section 1, Theorem 1.1, p. 4]. For $L = \log \mathcal{N}$ and fixed A_0, write H1$^\star(C_{\mathrm{zf},1}, C_{\mathrm{zf},2}; A_1)$ at $\mathcal{N}$ for the following assertion: for every $k \le L^{A_0}$ and every primitive character χ^* of conductor $q^* \mid k\mathcal{N}$ with $q^* > L^{A_1}$,

$$L(\sigma + it, \chi^*) \neq 0 \quad \left(\sigma \ge 1 - C_{\mathrm{zf},1} \frac{\log L}{L}, \quad |t| \le L^{C_{\mathrm{zf},2}}\right). \tag{1.3}$$

For every fixed choice of these constants, (1.2) implies (1.3) at all sufficiently large $\mathcal{N}$, since $1 - C_{\mathrm{zf},1} \log L/L > 7/8$.

Theorem 1.1 (Goldbach shift)**.** *Assume* H2′ *and* H1⋆ *at every sufficiently large even* $\mathcal{N}$*, with the fixed constants chosen in Theorem 6.12 and Section 10. Then, as* $\mathcal{N} \to \infty$ *through even integers,*

$$\sum_{p<\mathcal{N}} \mu(\mathcal{N}-p) = o(\pi(\mathcal{N})), \qquad \#\{p<\mathcal{N} : \mu(\mathcal{N}-p)=\kappa\} = \left(\frac{A_{\mathcal{N}}}{2}+o(1)\right)\pi(\mathcal{N}) \quad (\kappa=\pm 1). \tag{1.4}$$

In particular these conclusions are deduced from H2′ *and* (1.2)*. The errors are uniform in even* $\mathcal{N}$ *and ineffective.*

Theorem 1.2 (Small radical)**.** *Assume* H2′*. For each fixed* $C>0$*, the conclusions* (1.4) *hold as* $\mathcal{N}\to\infty$ *through even integers satisfying* $\operatorname{rad}(\mathcal{N}) \le (\log \mathcal{N})^C$*, using Iwaniec's region* (6.51) *and the Burgess bound* (6.38)*. The errors are uniform in this class for fixed* C *and ineffective.*

Theorem 1.3 (Small shifts)**.** *Assume the arguments of LWW as written:* H2′ *and, in addition, Sections 2.3, 3, 5 and 6 [6, pp. 6–14, 39–47]. For each fixed* $C>0$*, uniformly in integers* $0<|h|\le(\log x)^C$*,*

$$\sum_{\substack{p\le x\\ p+h\ge 1}} \mu(p+h) = o(\pi(x)), \qquad \#\{p\le x : p+h\ge 1,\ \mu(p+h)=\kappa\} = \left(\frac{A_h}{2}+o(1)\right)\pi(x) \quad (\kappa=\pm 1). \tag{1.5}$$

The errors are ineffective.

Theorem 1.4 (Liouville versions)**.** *Under the respective hypotheses of Theorems 1.1, 1.2 and 1.3, their conclusions hold with* λ *in place of* μ *and each sign density replaced by* $1/2+o(1)$*. The domain* $p+h\ge 1$ *is retained in the small-shift statement.*

Background. In their preprint submitted on 14 October 2026, Lü, Wang and Wu [6] state in Theorem 1 that $\sum_{p\le x}\mu(p+2)=o(\pi(x))$, and in Theorem 2 give the sign densities $A_2/2$ among all primes. Their abstract describes the conjecture as "dating back at least to Hildebrand's work in 1989"; Section 1 attributes its appearance in print to Hildebrand's 1989 paper [6, p. 3]. Their Section 4.4 describes the construction of its auxiliary operators as similar to Sections 4.1–4.3 of OpenAI's *The Poisson–Dirichlet law for prime predecessors*, dated 24 September 2026 [9]; their Appendix A says that its root-counting proof follows the argument of that paper's Lemma 3.3, with an explicit error for arbitrary squarefree S [6, pp. 26, 47]. The author's related papers, *Shifted primes with a prescribed parity of the number of prime factors* [3] and *Sums of two rough numbers with prescribed Liouville signs* [2], are dated 9 October 2026 in their repository entries.

What is used from LWW. We use Proposition 2.1 (the sifted Vaughan decomposition), Proposition 2.2 with (2.5) and (2.11) (twisted sifted Möbius sums), Proposition 2.3 (weight removal), Proposition 3.1 with (3.1), (3.4), (3.5) (Type I and Bombieri–Vinogradov), Lemma 4.1 with (4.6), and Proposition 6.1 with (6.3), as numbered inputs [6, pp. 4–16, 46–47]. We rework the expansion (4.3)–(4.5), the determinant equation (4.8), the block geometry, and the root-counting modifications of Lemma A.1 and (A.5). The arguments of Sections 4.2–4.7 and Appendix A are used as written after the changes in Appendix A. In particular the proof of Proposition 4.2, including Propositions 4.3–4.5, is the input H2′. The small-shift theorem also uses Lemmas 5.1–5.2 and the proof of Proposition 5.3 [6, pp. 40–45]; its assumption therefore includes more than H2′. Formula (5.15) [6, p. 44] is used only in Remark 6.17.

From *The Quasi-Riemann Hypothesis*, dated 30 September 2026, we use the zero-free half-plane (1.2). Its implication for (1.3) supplies an analytic input to Theorem 1.1. Remark 6.17 records a

separate consequence of (1.2) at polynomial height. For the small-radical class we use Iwaniec's Theorem 2 (1974, p. 97), stated in Input 6.13, and the $r = 2$ Burgess bound for arbitrary moduli in the form of Heath-Brown's introduction, equation (1) [4]. The real-character exception is removed in Lemma 6.14. The pointwise prime estimate needed here is established in Lemma 6.2, rather than obtained from a short-interval prime theorem.

Changes to LWW's argument. The determinant congruence decomposes the endpoint support into blocks on the unit-coordinate chart of $\mathbb{P}^1(\mathbb{Z}/\mathcal{N}\mathbb{Z})$. In a basis of each index-$\mathcal{N}$ lattice, $\det(g_{\mathcal{N}}z, g_{\mathcal{N}}z') = \mathcal{N}\det(z, z')$. The matrix replacing $\mathrm{diag}(1, -2)$ is $E_{\mathcal{N}} = \mathrm{diag}(1, \mathcal{N})$, followed by an integral shear. Lemma 4.1 gives the root count with both coordinates units modulo $\mathcal{N}$ and with the normalized complementary ratio. The major arcs reduce to the character-averaged energy (α^*) at polylogarithmic height. In Type I, $g = (r, \mathcal{N})$ is included in the extracted divisor before applying the Möbius identity, giving a reduced class modulo $2qr/g$.

Dependence and parameters. All uses of the LWW moment, memory, truncation and distinctness arguments have exactly the scope specified above. The obstruction in Proposition 6.18 explains a limitation of the classical zero-free inputs for this method; the region (1.3) is a sufficient input. We choose every precision and exponent before the large-parameter limit. The coefficient-loss constant C_{coef} and the zero-free constants $C_{\mathrm{zf},1}, C_{\mathrm{zf},2}$ are separate. For weight removal the large parameter tends to infinity with a fixed number b of disjoint weights, and then b tends to infinity. Every $o(1)$ here is ineffective through Siegel–Walfisz or LWW Proposition 2.2.

Outline. Sections 2–5 give the determinant reduction, blocks, root count and transport. Section 6 proves the major-arc energy estimate. Sections 7–10 complete the Goldbach statements. Sections 11 and 12 give the small-shift and Liouville statements. Appendix A records the changes to LWW by section, equation and page.

2 Setting and determinant reduction

Definition 2.1 (Parameters and coefficients)**.** For the Goldbach shift, $\mathcal{N}$ is even and $L = \log \mathcal{N}$. Fix $B > 2$, let $p \in (x, 2x]$, and put

$$\begin{aligned} &\frac{\mathcal{N}}{2L^B} \le x \le \frac{\mathcal{N}}{2}, \quad Z = \exp(L^{0.24}), \quad U = V = x^{1/5}, \\ &X = MN \asymp x, \quad M, N \ge x^{1/5}/2, \quad MN \le 2x \le \mathcal{N}. \end{aligned} \tag{2.1}$$

The letter N always denotes the length of the Vaughan variable n. Write $R_Z(n)$ for the indicator that every prime divisor of n exceeds Z. Fix $0 < \rho < 1$ and

$$\frac{1}{10} < a_1 < \cdots < a_K < \frac{1}{5}, \quad \mathcal{P}_i = \{p : \exp(L^{a_i}) < p \le \exp(2L^{a_i}),\ p \nmid \mathcal{N}\}, \quad V_i = \sum_{p \in \mathcal{P}_i} \frac{1}{p}. \tag{2.2}$$

Put $\mathcal{A} = \{p_1 \cdots p_K : p_i \in \mathcal{P}_i\}$, $\omega_i(n) = \#\{p \in \mathcal{P}_i : p \mid n\}$ and $\omega(n) = \sum_i \omega_i(n)$. The weight of LWW (2.13), p. 7, is

$$W(n) = \prod_{i=1}^{K} \frac{\rho^{\omega_i(n)-1}\omega_i(n)}{V_i}. \tag{2.3}$$

Let $J = \lfloor L^{0.01} \rfloor$, $R = \lceil L^{1/2}/2 \rceil$, and let d_0, D have the meanings of LWW Section 4.2, pp. 17–21. The real smooth dyadic cutoff η is supported on $[1,4]$ (LWW Section 4.1, p. 15), so its product scale Y has $a, b \in [Y, 4Y] \subset [Y, 16Y]$. Define

$$\Omega = [d_0 Y M, 16 d_0 Y M] \times [N, 2N]. \tag{2.4}$$

For $|v| \le L^{2B}$, the coefficients are

$$\begin{aligned} \alpha_m &= \mu(m) R_Z(m) \mathbf{1}_{(m,\mathcal{N})=1} m^{iv}, \\ \beta_n &= c_V(n) R_Z(n) \mathbf{1}_{(n,\mathcal{N})=1} n^{iv}, \qquad c_V(n) = \sum_{\substack{\ell \mid n \\ \ell > V}} \Lambda(\ell). \end{aligned} \tag{2.5}$$

Thus $|\alpha_m| \le 1$, $|\beta_n| \le L$, and the untwisted coefficient $c_V(n) R_Z(n) \mathbf{1}_{(n,\mathcal{N})=1}$ is nonnegative. For a character ξ of modulus dividing $k\mathcal{N}$, put

$$\begin{aligned} B_\xi(t) &= \frac{1}{Y \prod_i V_i} \sum_{b \in \mathcal{A}} \eta(b/Y) e(\lambda_0 b/Y) \xi(b) b^{it}, \\ M_\xi(t) &= \frac{1}{M} \sum_{M < m \le 2M} \alpha_m \xi(m) m^{it}, \qquad N_\xi(t) = \frac{1}{N} \sum_{N < n \le 2N} \beta_n \xi(n) n^{it}. \end{aligned} \tag{2.6}$$

Here $e(t) = \exp(2\pi i t)$, $k \le L^{A_0}$, $|\lambda_0| \le 2L^{A_0}$, $T_0 = \lceil L^{B_1} \rceil$, and $|B_\xi(t)| \le C_B = 16\|\eta\|_\infty / \prod_i V_i$. We write the arc displacement as λ_0 to distinguish it from the Liouville function. The major-arc target, at a fixed requested precision A, is

$$\frac{1}{\varphi(\mathcal{N})} \sum_{\chi \bmod \mathcal{N}} \int_{|t| \le T_0} |B_{\chi\psi_2}(t) M_{\chi\psi}(t) N_{\chi\psi}(t)|^2 \, dt \ll_A \frac{L^{-A}}{\mathcal{N}}. \tag{α^*}$$

The fixed twists ψ, ψ_2 are those constructed in Section 6.1. The trivial bound is $L^{O(1)}/\mathcal{N}$; $1/\mathcal{N}$ is the normalization.

Remark 2.2 (Ranges). Primes $p \le \mathcal{N}/L^B$ contribute $O(\mathcal{N}/L^B)$ to the bounded target sum, and $O(\mathcal{N} L^{1-B})$ with $\log p$. The rest splits into $O(\log L)$ dyadic intervals in (2.1). Before separating the product cutoff, $\mathcal{N} - mn \in [\mathcal{N} - 2x, \mathcal{N} - x) \cap (0, \mathcal{N})$. For fixed $a \asymp Y$ the positive cofactor belongs to

$$\left[\frac{\mathcal{N} - 2x}{a}, \frac{\mathcal{N} - x}{a} \right], \tag{2.8}$$

an interval of length $x/a \asymp X/Y$. As a varies, its centre varies, and the common positive support has $O(\mathcal{N}/Y)$ integers. Terms with $0 < \mathcal{N} - mn < Y$ vanish, since W requires a factor from every band. No dyadic condition on $\mathcal{N} - p$ is needed.

Proposition 2.3 (Vaughan reduction)**.** *Define $F(n) = \mu(\mathcal{N} - n) W(\mathcal{N} - n)$ for $1 \le n < \mathcal{N}$ and $F(n) = 0$ otherwise. LWW Proposition 2.1 [6, p. 4] applies to this bounded function. Its Type II sum is*

$$T_3(F) = \sum_{\substack{m > U, n > V \\ x < mn \le 2x}} \mu(m) R_Z(m) c_V(n) R_Z(n) \mu(\mathcal{N} - mn) W(\mathcal{N} - mn). \tag{2.9}$$

The smoothing and Mellin separation in LWW Section 4.1, p. 14, reduce it to rectangular sums

$$T = \sum_{m,n} \alpha_m \beta_n F(mn), \qquad M, N \ge x^{1/5}/2, \quad MN \asymp x, \quad |v| \le L^{2B}, \tag{2.10}$$

with the coprimality restrictions justified below. The argument of μ is positive wherever F is nonzero.

Proof. Apply the stated decomposition and separate the cutoff in mn/x. It concerns the prime variable, so this separation has the same Mellin norm and coefficient bounds for F as in LWW. On parts of a rectangle where $mn \geq \mathcal{N}$, the definition $F(mn) = 0$ applies. □

Lemma 2.4 (Puncturing and unit support)**.** *The punctured bands satisfy $V_i = \log 2 + o(1)$, uniformly in $\mathcal{N}$. The restrictions $(m, \mathcal{N}) = (n, \mathcal{N}) = 1$ in (2.5) cost $O_A(XL^{-A})$ for every fixed A. The remaining variables are units modulo $k\mathcal{N}$ whenever $k \leq L^{A_0}$.*

Proof. The removed reciprocal mass is at most

$$\sum_{\substack{p \mid \mathcal{N} \\ p > \exp(L^{0.1})}} \frac{1}{p} \leq \omega_{\mathrm{all}}(\mathcal{N}) e^{-L^{0.1}} \leq L e^{-L^{0.1}}, \tag{2.11}$$

where ω_{all} counts all distinct prime divisors. Thus LWW (2.14)–(2.15), p. 7, apply with the same limits. A common prime of a sifted variable and $\mathcal{N}$ exceeds Z. The number of discarded pairs is bounded, up to an absolute constant, by

$$\sum_{\substack{p \mid \mathcal{N} \\ p > Z}} \frac{X}{p} \leq \frac{X \omega_{\mathrm{all}}(\mathcal{N})}{Z} \leq XLe^{-L^{0.24}}. \tag{2.12}$$

The bound $|\alpha_m \beta_n| \ll L$ absorbs this into the claimed error. The restrictions preserve the coefficient and sifting hypotheses of Proposition 4.2. Every prime factor of k is below Z, so the sifted variables are also units modulo k. Every list prime and every $a, b \in \mathcal{A}$ is a unit modulo $\mathcal{N}$ by puncturing, and modulo k because the bands lie above L^{A_0}. □

Proposition 2.5 (The cofactor square)**.** *After a smooth dyadic partition $\eta(a/Y)$, the expansion of (2.10) has the form*

$$T = \frac{(-1)^K}{\prod_i V_i} \sum_{h \geq 1} \mu(h) \rho^{\omega(h)} \sum_{\substack{p_i \in \mathcal{P}_i, m, n \\ mn + ah = \mathcal{N}}} \alpha_m \beta_n + O(XLe^{-L^{0.1}}). \tag{2.13}$$

Its dyadic pieces satisfy

$$|T_Y|^2 \ll \frac{\mathcal{N}}{Y} Q_Y^+ \leq \frac{\mathcal{N}}{Y} Q_Y^{\mathrm{all}}, \qquad Q_Y^{\mathrm{all}} = \sum_{\substack{h \in \mathbb{Z} \\ h \neq 0}} \left| \frac{1}{\prod_i V_i} \sum_{\substack{a \in \mathcal{A}, m, n \\ \mathcal{N} - mn = ah}} \eta(a/Y) \alpha_m \beta_n \right|^2 \geq 0. \tag{2.14}$$

Here Q_Y^+ is the same square restricted to $h \geq 1$. All cofactors in Q_Y^{all} satisfy $|h| \ll \mathcal{N}/Y$.

Proof. If $\mathcal{N} - mn = ah$ and $(a, h) = 1$, then $\mu(ah) = (-1)^K \mu(h)$ and $\omega(ah) = K + \omega(h)$. Also $(a, m) = 1$: a common list prime would divide $\mathcal{N}$. When $(a, h) > 1$, some list prime has square dividing $\mathcal{N} - mn$. For fixed m this is one residue class of n modulo that square, and the count in LWW (4.3), p. 15, gives $O(XLe^{-L^{0.1}})$. The coprimality restriction can be removed in the expansion at the same cost; its new terms have that square-divisibility property. Cauchy–Schwarz uses $|\mu(h)\rho^{\omega(h)}| \leq 1$ and the common range $O(\mathcal{N}/Y)$ from Remark 2.2. Each square is nonnegative, proving $Q_Y^+ \leq Q_Y^{\mathrm{all}}$. On a supported rectangle $mn \ll MN \leq \mathcal{N}$, whence $|\mathcal{N} - mn| \ll \mathcal{N}$. The cofactor 0 is impossible because $mn = \mathcal{N}$ contradicts $(mn, \mathcal{N}) = 1$ for $\mathcal{N} > 1$. Only this upper-bound quadratic form introduces negative cofactors; F and the expansion of its target always use positive arguments. □

Lemma 2.6 (Disjoint lists and determinant equation)**.** *Terms with $(a,b)>1$ may be removed from Q_Y^{all} at a cost $O_A(\mathcal{N}YL^{-A})$. For $(a,b)=1$ the exact determinant equation is*

$$b(\mathcal{N}-mn)=a(\mathcal{N}-rs) \quad\Longleftrightarrow\quad t:=bmn-ars=\mathcal{N}(b-a). \tag{2.15}$$

Equality on the left is equality of the two integer cofactors, with no positivity condition. In particular $\mathcal{N}\mid t$ and $|t|\ll\mathcal{N}Y$, while $t'=t/\mathcal{N}=b-a$ has size $O(Y)$.

Proof. The divisor-square estimate used in LWW Lemma 4.1 and (4.6), p. 16, has the form

$$\sum_{h\in I_\ell}\tau(\mathcal{N}-\ell h)^2\ll(X/Y)L^3,\qquad |I_\ell|\ll X/Y \tag{2.16}$$

where I_ℓ is the translated cofactor interval for fixed $\ell\asymp Y$, with $\mathcal{N}-\ell h$ in the positive dyadic product range. The argument depends on $d\mid\mathcal{N}-\ell h\Rightarrow(d,\ell)\mid\mathcal{N}$; since $(\ell,\mathcal{N})=1$, it gives $(d,\ell)=1$, exactly the needed hypothesis. For the all-integer square use the full interval $|h|\ll\mathcal{N}/Y$ and $|\mathcal{N}-\ell h|$ in its absolute counts. Its length costs at most $\mathcal{N}/X\ll L^B$ relative to the displayed local estimate. LWW's divisor argument and the common-list saving then give the stated $\mathcal{N}YL^{-A}$ after increasing the fixed precision. Reducing (2.15) modulo a and b, and using $(a,b)=1$, proves divisibility by a,b of the corresponding numerators and equality of their integer quotients. □

3 Blocks and relative coordinates

Endpoint positions, with the list data of LWW attached, are

$$P=(Dbm,s)^T,\qquad Q=(Dar,n)^T,\qquad \det(P,Q)=D(bmn-ars)=Dt. \tag{3.1}$$

All endpoint coordinates, and D,a,b, are units modulo $\mathcal{N}$.

3.1 The major-arc congruence

Let $\mathfrak{M}_1$ be LWW's union of arcs with denominators $k\le L^{A_0}$ and width $2L^{A_0}/Y$ [6, Section 4.1, p. 16]. Define

$$\begin{aligned}\mathfrak{M}_{\mathcal{N}}&=\{\theta\in\mathbb{T}:\mathcal{N}\theta\in\mathfrak{M}_1\},\\ H_{\mathcal{N}}(t;a,b)&=\psi_1(t/(\mathcal{N}Y))\int_{\mathfrak{M}_{\mathcal{N}}}e(\theta(t-\mathcal{N}(b-a)))\,d\theta,\\ Q_Y^{\mathrm{maj},\mathcal{N}}&=\prod_i V_i^{-2}\sum_{a,b,m,n,r,s}\eta(a/Y)\eta(b/Y)\alpha_m\overline{\alpha_r}\beta_n\overline{\beta_s}H_{\mathcal{N}}(bmn-ars;a,b).\end{aligned} \tag{3.2}$$

The cutoff ψ_1 is the fixed smooth cutoff of LWW, equal to 1 on $[-4,4]$.

Lemma 3.1 (Covering map)**.** *For every integer t,*

$$\int_{\mathfrak{M}_{\mathcal{N}}}e(\theta(t-\mathcal{N}(b-a)))\,d\theta=\mathbf{1}_{\mathcal{N}\mid t}\int_{\mathfrak{M}_1}e(\phi(t/\mathcal{N}-b+a))\,d\phi. \tag{3.3}$$

In particular $|\mathfrak{M}_{\mathcal{N}}|=|\mathfrak{M}_1|\ll L^{3A_0}/Y$.

Proof. For a fundamental domain of $\mathfrak{M}_1$, the change of variables gives

$$\int_{\mathfrak{M}_{\mathcal{N}}} F(\theta)\,d\theta = \frac{1}{\mathcal{N}} \sum_{j \bmod \mathcal{N}} \int_{\mathfrak{M}_1} F((\phi+j)/\mathcal{N})\,d\phi.$$

For the integrand in (3.3),

$$F((\phi+j)/\mathcal{N}) = e(\phi t/\mathcal{N})e(jt/\mathcal{N})e(-\phi(b-a))e(-j(b-a)).$$

The last factor is 1 and the j-sum is $\mathcal{N}\mathbf{1}_{\mathcal{N}|t}$. This also makes the identity independent of the chosen representatives. □

The raw-minus-major kernel is

$$\begin{aligned} K_{\mathcal{N}}(P,Q) = \frac{d_0}{D}\eta(a/Y)\eta(b/Y)\psi_1(t/(\mathcal{N}Y)) \\ \cdot \left\{ \mathbf{1}_{t=\mathcal{N}(b-a)} - \int_{\mathfrak{M}_{\mathcal{N}}} e(\theta(t-\mathcal{N}(b-a)))\,d\theta \right\} \rho^{(\omega(P_1)+\omega(Q_1)-2K(J+1))/2}. \end{aligned} \tag{3.4}$$

Both terms vanish unless $\mathcal{N} \mid t$. On that congruence the brace is

$$\mathbf{1}_{t'=b-a} - \int_{\mathfrak{M}_1} e(\phi(t'-b+a))\,d\phi, \qquad t' = t/\mathcal{N}, \tag{3.5}$$

with cutoff $\psi_1(t'/Y)$, the transformed kernel of LWW Section 4.2, p. 18.

3.2 Lines and their sizes

Definition 3.2 (Unit endpoints and labelled blocks)**.** Let $\mathcal{C}_{\mathcal{N}}$ consist of $P \in \Omega$ primitive in $\mathbb{Z}^2$ with $(P_1P_2, \mathcal{N}) = 1$. For a unit $\varrho \bmod \mathcal{N}$, put

$$\Lambda_\varrho = \{(u,v) \in \mathbb{Z}^2 : v \equiv \varrho u \pmod{\mathcal{N}}\}. \tag{3.6}$$

This has basis $(1,\varrho), (0,\mathcal{N})$ and index $\mathcal{N}$. The slope class of a unit-coordinate endpoint is $\varrho(P) = P_2\overline{P_1} \bmod \mathcal{N}$.

Lemma 3.3 (Block diagonality)**.** *For unit-coordinate endpoints,*

$$\mathcal{N} \mid \det(P,Q) \iff \varrho(P) = \varrho(Q), \qquad \mathcal{C}_{\mathcal{N}} = \coprod_{\varrho \in (\mathbb{Z}/\mathcal{N}\mathbb{Z})^\times} (\mathcal{C}_{\mathcal{N}} \cap \Lambda_\varrho). \tag{3.7}$$

Thus the endpoint kernel is block diagonal. Each full lattice is regarded as a labelled copy when the state space is enlarged.

Proof. Invert P_1Q_1 in the determinant congruence. Since $(D,\mathcal{N}) = 1$, $\mathcal{N} \mid t$ is equivalent to $\mathcal{N} \mid \det(P,Q)$. The unit restriction makes the slope class unique. □

Lemma 3.4 (Block size)**.** *The number of vectors in $\Omega \cap \Lambda_\varrho$ primitive in Λ_ϱ is*

$$O\left(1 + \frac{d_0 YMN}{\mathcal{N}}\right). \tag{3.8}$$

In the application $d_0YMN/\mathcal{N} \to \infty$ and $MN \le \mathcal{N}$, so this is $O(d_0YMN/\mathcal{N}) \le O(d_0Y)$, uniformly in ϱ.

Proof. If the block is occupied, choose a lattice-primitive P_0 in it. For any Q in the block,

$$\det(P_0, Q) \in \mathcal{N}\mathbb{Z}, \qquad |\det(P_0, Q)| \le 2 \cdot 16 d_0 Y M \cdot 2N = 64 d_0 Y M N \le 64 d_0 Y \mathcal{N}.$$

There are $O(1 + d_0 Y M N/\mathcal{N})$ possible determinants. For each, two solutions differ by an element of $\mathbb{Q}P_0 \cap \Lambda_\varrho = \mathbb{Z}P_0$; their first coordinates permit at most 16 solutions. Here $\mathbb{Q}$ denotes the rational numbers. In the application $\mathcal{N}/(d_0 Y M N) \ll L^B/(d_0 Y) \to 0$. □

This partition supplies pieces of the size of LWW's slope intervals [6, Section 4.6, p. 36]. Physical slope locality is replaced by relative-coordinate bounds.

3.3 A normalized basis

Let $P = (u, v)^T \in \mathcal{C}_\mathcal{N}$, and choose $ud - vc = 1$, $0 \le c < u$. Define

$$\begin{aligned} &c' \equiv \mathcal{N} c \pmod u, \quad 0 \le c' < u, \qquad c'v \equiv -\mathcal{N} \pmod u, \quad \beta_P = c'/u, \\ &d'' = (\mathcal{N} + c'v)/u, \qquad P^* = (c', d'')^T, \qquad g_\mathcal{N}(P) = \begin{pmatrix} u & c' \\ v & d'' \end{pmatrix}, \quad \det g_\mathcal{N}(P) = \mathcal{N}. \end{aligned} \tag{3.9}$$

Equivalently

$$P^* = \mathcal{N}(c, d)^T - \lfloor \mathcal{N} c/u \rfloor P, \qquad g_\mathcal{N}(P) = g E_\mathcal{N} U_0, \quad g \in \mathrm{SL}_2(\mathbb{Z}), \quad E_\mathcal{N} = \mathrm{diag}(1, \mathcal{N}), \tag{3.10}$$

where U_0 is the corresponding unimodular shear.

Lemma 3.5 (Relative coordinates)**.** *Writing $\varrho = \varrho(P)$ and $Q = g_\mathcal{N}(P)z$, one has:*

1. $\Lambda_\varrho = \mathbb{Z}P \oplus \mathbb{Z}P^*$.
2. *With $\tau_\mathcal{N}(z) = z_1 + \beta_P z_2$,*

$$\det(P, Q) = \mathcal{N} z_2, \quad Q_1 = u\tau_\mathcal{N}(z), \quad Q_2 = v\tau_\mathcal{N}(z) + \frac{\mathcal{N} z_2}{u}. \tag{3.11}$$

3. *Q is primitive in Λ_ϱ precisely when $(z_1, z_2) = 1$. It is primitive in $\mathbb{Z}^2$ precisely when also $(Q_1, \mathcal{N}) = 1$.*
4. *For primitive z and every prime $p \nmid \mathcal{N}$,*

$$p \mid Q_1 \quad \Longleftrightarrow \quad [z]_p = \ker(u, c') \in \mathbb{P}^1(\mathbb{F}_p). \tag{3.12}$$

5. *For $Q \in \Omega$,*

$$|z_2| \le \frac{64 d_0 Y M N}{\mathcal{N}} \le 64 d_0 Y, \qquad |z_1| \le 16 + |z_2| \le 2^7 d_0 Y. \tag{3.13}$$

Conversely every integer z gives $Q \in \Lambda_\varrho$.

Proof. The two columns belong to Λ_ϱ and have its determinant. Formula (3.11) follows from $d'' = \mathcal{N}/u + \beta_P v$. The lattice bijection proves lattice-primitivity. At $p \mid \mathcal{N}$ the matrix has rank one, with unit-coordinate column P, and $Q \equiv 0 \pmod p$ precisely when $p \mid Q_1$. At $p \nmid \mathcal{N}$ the matrix is invertible, so $Q \equiv 0$ precisely when $z \equiv 0$. Also $(u, c') = 1$, since $c'v \equiv -\mathcal{N} \pmod u$ and $(u, \mathcal{N}) = 1$. This proves the kernel assertion for primitive z. Finally use the determinant bound in Lemma 3.4, $Q_1/u \le 16$ and $0 \le \beta_P < 1$. □

The continuous extension and its comparison with LWW are

$$g_{\mathcal{N}}(u,v,\beta)=\begin{pmatrix} u & \beta u \\ v & \mathcal{N}/u+\beta v\end{pmatrix},\quad \tau_{\mathcal{N}}(z)=z_1+\beta z_2,\quad g'(u,v,r)=\begin{pmatrix} u & -2ur \\ (u-v)/2 & r(v-u)+1/u\end{pmatrix}. \tag{3.14}$$

Thus $-2r$ is replaced by β and $1/u$ by $\mathcal{N}/u$; both entries are retained exactly. The ratio in LWW (4.26), p. 21, becomes $\beta_P=\{\mathcal{N}r_P\}$. Multiplication by $\mathcal{N}$ preserves Lebesgue measure on $\mathbb{T}$.

3.4 Enlarged states and intrinsic ratios

Definition 3.6 (States and good-state ratios)**.** The ambient position space is the disjoint union of labelled sets

$$\mathcal{S}'_\varrho=\{(\varrho,Q): Q\in\Omega\cap\Lambda_\varrho,\ Q \text{ primitive in } \Lambda_\varrho\}, \tag{3.15}$$

with LWW's lists attached. For such a state choose $W\in\Lambda_\varrho$ with $\det(Q,W)=\mathcal{N}$ and put

$$\beta_Q=\frac{W_1}{Q_1}\pmod 1. \tag{3.16}$$

The good-state tests of LWW (4.14)–(4.16), pp. 18–19, use this ratio at every state.

Lemma 3.7 (Intrinsic ratio and endpoint compression)**.** *Definition 3.6 is independent of the complement and of the chosen block basis. In relative coordinates,*

$$\beta_Q\equiv\frac{\tau_{\mathcal{N}}(w)}{\tau_{\mathcal{N}}(z)}\pmod 1\quad \big(Q=g_{\mathcal{N}}(P)z, W=g_{\mathcal{N}}(P)w, \det(z,w)=1\big). \tag{3.17}$$

For a unit endpoint it equals $\{\mathcal{N}r_Q\}$. LWW's test that z_1 is odd is replaced by endpoint support $(Q_1Q_2,\mathcal{N})=1$, which is not a parity statement about the basis used here. The restricted-root trace in LWW Section 4.6 is valid in this space.

Proof. A lattice-primitive Q has a determinant-$\mathcal{N}$ complement; any two such complements differ by an integer multiple of Q. Two oriented bases differ by $\mathrm{SL}_2(\mathbb{Z})$, proving (3.17) and basis independence. At a unit endpoint the normalized complement in (3.9) gives $\{\mathcal{N}r_Q\}$. A nonunit vector can belong to several full lattices, giving a different labelled state in each. Such a vector is not primitive in $\mathbb{Z}^2$, by Lemma 3.5; the endpoint vector vanishes there because its coefficients and lists are supported on units. For example, $\mathcal{N}=2$, $P=(3,1)$, $P^*=(1,1)$ and $z=(0,1)$ give a unit endpoint with z_1 even. Thus no assertion z_1 odd is made after this shear.

What the moment argument uses is a completion at a root, the primitive-coordinate equivalence

$$D\mid\det(z,z')\iff [z]_p=[z']_p\quad (p\mid D),$$

for squarefree D, and the ratio at the current state. All hold here. The diagonal operators S,G preserve the endpoint pairing under enlargement. For positive B_0, the matrix

$$\big(\langle u_P,B_0^R u_Q\rangle\big)_{P,Q\in\mathcal{C}_{\mathcal{N}}\cap\Lambda_\varrho}\succeq 0,\qquad \lambda_{\max}\le d_\varrho:=\sum_{P\in\mathcal{C}_{\mathcal{N}}\cap\Lambda_\varrho}\langle u_P,B_0^R u_P\rangle \tag{3.18}$$

is a principal submatrix of a Gram matrix. Spectral Hölder in LWW (4.65), p. 36, is applied in the ambient space before this restriction. The sum $\sum_\varrho d_\varrho$ is exactly the root average over $\Omega\cap\mathcal{C}_{\mathcal{N}}$ counted below. No estimate for the full trace on added nonunit states is required. The second good-state test in LWW (4.47)–(4.48), p. 30, is applied at the source position Q, using β_Q. For the exceptional-measure bound, multiplication by $\mathcal{N}$ preserves Haar measure; for the subsequent cube count we count β directly. □

4 Root counting

Lemma 4.1 (Lemma R). *Put $U_0 = d_0 Y M$, $V_0 = N$, and fix $\delta > 0$. Suppose $U_0, V_0 \ge x^\delta$, $\mathcal{N} < U_0 V_0$, and $\log x = L + O_B(\log L)$. Let S be odd and squarefree, $(S, \mathcal{N}) = 1$, $S \le \exp(C_0 L^{0.98})$, and let $g_0 \bmod S$ have determinant $\mathcal{N} \bmod S$. For intervals $I_1 \subset [1,16]$, $I_2 \subset [1,2]$ and $I_3 \subset [0,1]$,*

$$\begin{aligned}\#\{P = (u,v)^T : (u,v) = 1,\ (uv, \mathcal{N}) = 1,\ u/U_0 \in I_1,\ v/V_0 \in I_2,\ \beta_P \in I_3,\ g_{\mathcal{N}}(P) \equiv g_0 \pmod S\}\\ = \frac{c_{\mathcal{N}} U_0 V_0 |I_1||I_2||I_3|}{|\operatorname{SL}_2(\mathbb{Z}/S\mathbb{Z})|} + O(U_0 V_0 x^{-\delta/800}),\end{aligned} \tag{4.1}$$

where

$$c_{\mathcal{N}} = \frac{1}{\zeta(2)} \prod_{p \mid \mathcal{N}} \frac{p-1}{p+1} = \frac{1}{\zeta(2)} \prod_{p \mid \mathcal{N}} \frac{(1-1/p)^2}{1-1/p^2}, \qquad (\log\log \mathcal{N})^{-2} \ll c_{\mathcal{N}} \le 1. \tag{4.2}$$

For prescribed lines $\ell_p \in \mathbb{P}^1(\mathbb{F}_p)$ at $p \mid S$,

$$\begin{aligned}(U_0 V_0)^{-1} \#\{P : (u,v) = 1, (uv, \mathcal{N}) = 1,\ u/U_0 \in I_1,\ v/V_0 \in I_2,\\ \beta_P \in I_3,\ \ker(g_{\mathcal{N}}(P)_{1,*} \bmod p) = \ell_p\ (p \mid S)\}\\ = c_{\mathcal{N}} |I_1||I_2||I_3| \prod_{p \mid S} \frac{1}{p+1} + O(x^{-c\delta})\end{aligned} \tag{4.3}$$

for an absolute $c > 0$, possibly smaller than $1/800$. All errors are uniform in the intervals, matrix class and $\mathcal{N}$ in the stated dyadic regime.

Proof. We use LWW Appendix A, (A.2)–(A.4), pp. 48–50, and give the changes to Lemma A.1 and (A.5), pp. 47–51.

The matrix and the shorter-coordinate case. The matrix (3.9) has determinant $\mathcal{N}$ and $0 \le c' < u$. Suppose $U_0 \le V_0$, and fix $u \equiv u_0 \pmod S$ with $(u, \mathcal{N}) = 1$. Replace LWW's set by

$$\mathcal{R}_u = \{(c', v) \bmod uS : c' \equiv c'_0\ (S),\ v \equiv v_0\ (S),\ c'v \equiv u d''_0 - \mathcal{N}\ (uS)\}. \tag{4.4}$$

The right side in the last congruence is a unit at every prime dividing u. At primes dividing both u and S this follows also from $u_0 d''_0 - c'_0 v_0 \equiv \mathcal{N} \not\equiv 0 \pmod p$. Consequently the same count as in LWW gives

$$\#\mathcal{R}_u = u \prod_{\substack{p \mid u \\ p \nmid S}} (1 - 1/p), \qquad (u, c') = (u, v) = 1. \tag{4.5}$$

Exponential sums. Let $S_1 = (u, S)$, $S_2 = S/S_1$, and $\gamma S_2 \equiv 1 \pmod{uS_1}$. The CRT calculation of LWW (A.3), p. 49, gives

$$\left| \sum_{(c',v) \in \mathcal{R}_u} e\left(\frac{hc' + kv}{uS}\right) \right| \le \frac{1}{S_1} \sum_{j \bmod S_1} |K_{uS_1}(\gamma h + ju, \gamma k(u d''_0 - \mathcal{N}))|. \tag{4.6}$$

Here $K_m(h,k) = \sum_{y \bmod m, (y,m)=1} e((hy + k\bar{y})/m)$. Since $S_1 \mid u$ and $(u, \mathcal{N}) = 1$, $u d''_0 - \mathcal{N}$ is a unit modulo uS_1. Thus

$$(\gamma h + ju, \gamma k(u d''_0 - \mathcal{N}), uS_1) \le S_1(h, k, u).$$

The all-modulus fourth-moment bound (A.2), followed by (A.3) and Fourier completion (A.4), gives rectangle discrepancy $O(u^{0.99}S^2)$, independent of the rectangle. No prime-modulus Kloosterman estimate is used.

The sieve on v. Expand $\mathbf{1}_{(v,\mathcal{N})=1} = \sum_{d|\mathrm{rad}(\mathcal{N})} \mu(d)\mathbf{1}_{d|v}$. There are at most $2^{\omega_{\mathrm{all}}(\mathcal{N})} = x^{o(1)}$ terms. Every d is coprime to uS; replacing v by dv_2 replaces the right side of (4.4) by the unit $\bar{d}(ud_0'' - \mathcal{N})$ and the prescribed residue by $v_2 \equiv \bar{d}v_0 \pmod{S}$. Therefore the same rectangle error applies. Splitting into periods as in LWW Appendix A, p. 50, the count for this d is

$$\frac{V_0|I_2||I_3|}{dS^2}\prod_{\substack{p|u\\ p\nmid S}}(1-1/p) + O\left(\left(\frac{V_0}{uSd}+1\right)u^{0.99}S^2\right). \tag{4.7}$$

The sum with $\mu(d)$ multiplies the main term by $\prod_{p|\mathcal{N}}(1-1/p)$ and leaves error $O(x^{o(1)}(V_0/(uS)+1)u^{0.99}S^2)$.

The sieve on u and the local constant. Use

$$\prod_{\substack{p|u\\ p\nmid S}}(1-1/p) = \sum_{\substack{\ell|u\\ (\ell,S)=1}}\frac{\mu(\ell)}{\ell}, \qquad \mathbf{1}_{(u,\mathcal{N})=1} = \sum_{d|\mathrm{rad}(\mathcal{N})}\mu(d)\mathbf{1}_{d|u}.$$

For $(\ell d, S) = 1$, counting $u \in U_0I_1$ in its fixed residue class gives $U_0|I_1|/(\mathrm{lcm}(\ell,d)S) + O(1)$. The double divisor series has local factors

$$\begin{cases} 1-p^{-2}, & p\nmid S\mathcal{N},\\ (1-p^{-2})-p^{-1}(1-p^{-1}) = 1-p^{-1}, & p\mid\mathcal{N},\\ 1, & p\mid S. \end{cases} \tag{4.8}$$

To justify its tail without a factor $2^{\omega_{\mathrm{all}}(\mathcal{N})}$, sum over d before extending ℓ:

$$\sum_{d|\mathrm{rad}(\mathcal{N})}\frac{\mu(d)}{\mathrm{lcm}(\ell,d)} = \begin{cases} 0, & (\ell,\mathcal{N})>1,\\ \ell^{-1}\prod_{p|\mathcal{N}}(1-1/p), & (\ell,\mathcal{N})=1. \end{cases} \tag{4.9}$$

Indeed a prime common to ℓ and $\mathcal{N}$ makes the two d-choices cancel; otherwise the choices contribute $1-1/p$. After the factor $\mu(\ell)/\ell$, the tail $\ell > 16U_0$ is $O(\sum_{\ell>16U_0}\ell^{-2}) = O(1/U_0)$. The integer-counting errors total $O(2^{\omega_{\mathrm{all}}(\mathcal{N})}\log U_0) = O(x^{o(1)})$. The resulting u main term is

$$\frac{U_0|I_1|}{S\zeta(2)}\prod_{p|S}(1-p^{-2})^{-1}\prod_{p|\mathcal{N}}(1+1/p)^{-1}. \tag{4.10}$$

Multiplication by the v-sieve factor and $|\mathrm{SL}_2(\mathbb{Z}/S\mathbb{Z})| = S^3\prod_{p|S}(1-p^{-2})$ gives (4.1) with (4.2). At $\mathcal{N}=2$ this is $1/(3\zeta(2))$. For completeness, split the factors $1-1/p$ at $p=L$. Mertens' product estimate gives $\prod_{p|\mathcal{N},p\le L}(1-1/p) \ge \prod_{p\le L}(1-1/p) \gg 1/\log L$. For $p > L \ge 2$, $\log(1-1/p) \ge -2/p$ and $\sum_{p|\mathcal{N},p>L} 1/p \le \omega_{\mathrm{all}}(\mathcal{N})/L \le 1/\log 2$. Thus the product over these larger primes is bounded below by an absolute positive constant. It follows from the squared-factor formula in (4.2) that

$$c_{\mathcal{N}} \ge \zeta(2)^{-1}\prod_{p|\mathcal{N}}(1-1/p)^2 \gg (\log L)^{-2} = (\log\log\mathcal{N})^{-2}. \tag{4.11}$$

The upper bound $c_{\mathcal{N}} \le 1$ is immediate. Only that upper bound is needed for the root moment.

Error for $U_0 \le V_0$. Summing the rectangle errors over $u \asymp U_0$ gives

$$\begin{aligned} O\big(x^{o(1)}S^2(V_0U_0^{0.99}+U_0^{1.99})\big) &= O\big(U_0V_0x^{o(1)}S^2(U_0^{-0.01}+U_0^{0.99}/V_0)\big) \\ &= O(U_0V_0x^{-c_0\delta}), \qquad c_0 = 1/200. \end{aligned} \tag{4.12}$$

Both terms in parentheses are at most $x^{-\delta/100}$ and $S^2 \le \exp(2C_0L^{0.98}) = x^{o(1)}$. This fixes c_0 before the next subdivision.

The case $V_0 < U_0$. Complete $(v,u)^T$ in the normalized way: $c''u \equiv -\mathcal{N} \pmod v$, $0 \le c'' < v$, $d''' = (\mathcal{N}+c''u)/v$. Put $\varepsilon = \mathcal{N}/(uv)$. Our hypothesis gives

$$0 < \varepsilon \le \frac{\mathcal{N}}{d_0YMN} < 1, \qquad \frac{\mathcal{N}}{d_0YMN} \ll \frac{L^B}{d_0Y} \quad \text{in the application.} \tag{4.13}$$

Since $d''/v = \beta_P + \varepsilon$, $k = \lfloor d''/v \rfloor \in \{0,1\}$, and

$$\begin{aligned} &c'' = (1+k)v - d'', \quad d''' = (1+k)u - c', \\ &\widetilde{g} = \begin{pmatrix} v & (1+k)v-d'' \\ u & (1+k)u-c' \end{pmatrix}, \quad \det\widetilde{g} = \mathcal{N}, \\ &\beta_P'' = 1+k-\beta_P-\varepsilon, \qquad k=1 \iff \beta_P+\varepsilon \ge 1. \end{aligned} \tag{4.14}$$

Either endpoint $\beta_P'' \in \{0,1\}$ would imply $v \mid \mathcal{N}$, impossible for $v > 1$ and $(v,\mathcal{N}) = 1$. Both k branches must be included. For $k = \kappa \in \{0,1\}$ the prescribed residue becomes

$$\widetilde{g}_0^{(\kappa)} = \begin{pmatrix} v_0 & (1+\kappa)v_0 - d_0'' \\ u_0 & (1+\kappa)u_0 - c_0' \end{pmatrix}, \quad J_\kappa(u,v) = \big((1+\kappa) - I_3 - \mathcal{N}/(uv)\big) \cap [0,1). \tag{4.15}$$

Their union is $1-(I_3+\mathcal{N}/(uv))$ modulo 1, of length $|I_3|$.

Partition $I_1 \times I_2$ into cells of relative side $x^{-\kappa_0}$, $\kappa_0 = c_0\delta/4$. There are $O(x^{c_0\delta/2})$ cells. On each, $\mathcal{N}/(uv)$ varies by at most $4x^{-\kappa_0}$. For each branch start with the affine interval $(1+\kappa)-I_3-\varepsilon_0$, where ε_0 is its corner value. Enlarge or contract this interval by $4x^{-\kappa_0}$ at each end and only then intersect with $[0,1)$, taking reversed intervals as empty. This also covers a branch whose clipped interval is empty at the corner. Apply the shorter-coordinate case on that cell, with the two coordinate roles interchanged and residue $\widetilde{g}_0^{(\kappa)}$. Its error is independent of interval lengths, so the summed error is $O(U_0V_0x^{-c_0\delta/2})$. The boundary mass is at most $8x^{-\kappa_0}$ per cell times its main-term density, and totals $O(U_0V_0x^{-\kappa_0})$. The two branches preserve $|I_3|$, giving final exponent $c_0/4 = 1/800$ in (4.1). This argument counts $\beta = c'/u$ directly. It does not differentiate the map $r \mapsto \mathcal{N}r$.

Prescribed lines. At $p \mid S$, matrices of determinant $\mathcal{N}$ with a prescribed first-row kernel number $p(p-1)$: there are $p-1$ nonzero first rows with that kernel and p choices of second row. The total number is $p(p^2-1)$, so the proportion is $1/(p+1)$. CRT multiplies these proportions. Summing the per-matrix errors costs at most $S^3 = x^{o(1)}$; decrease the fixed positive exponent c in (4.3), for example to $1/1600$ for sufficiently large x with fixed δ. The factor at 2 is already included in $c_{\mathcal{N}}$ and is not inserted again. □

5 Transport of the Type II argument

Lemma 5.1 (Reduced determinant and boundary geometry)**.** *For $Q = g_{\mathcal{N}}(P)z$, $Q' = g_{\mathcal{N}}(P)z'$ in one labelled block,*

$$\det(g_{\mathcal{N}}(P)z, g_{\mathcal{N}}(P)z') = \mathcal{N}\det(z,z'), \qquad t' = \frac{\det(Q,Q')}{\mathcal{N}D} = \frac{\det(z,z')}{D}. \tag{5.1}$$

The kernel in relative coordinates is (3.5) with cutoff $\psi_1(\det(z,z')/(DY))$. On allowed positions and edges,

$$\frac{1}{16} \le \tau_{\mathcal{N}}(z) \le 16, \qquad |\det(z,z')| \ll d_0 Y, \qquad |z| \ll d_0 Y M N/\mathcal{N} \le d_0 Y. \tag{5.2}$$

For $s = u/(d_0 Y M)$, $t_0 = v/N$, the position boundaries are

$$s\tau_{\mathcal{N}}(z) \in \{1, 16\}, \qquad E_2 = t_0 \tau_{\mathcal{N}}(z) + \frac{z_2 \mathcal{N}}{d_0 Y M N s} \in \{1, 2\}. \tag{5.3}$$

The second term is bounded by 64 in absolute value on allowed positions; its s derivative is $O(1)$, and $\partial_{t_0} E_2 = \tau_{\mathcal{N}}(z) \ge 1/16$.

Proof. The determinant identities follow from $\det g_{\mathcal{N}} = \mathcal{N}$ and (3.1). The remaining bounds follow from Lemma 3.5 and the cutoff in (3.4). In the last boundary formula $s \in [1, 16]$, so the extra term and its s derivative are $O(1)$; its β derivative is $t_0 z_2 = \exp(O(L^{0.21}))$. Transversality, rather than smallness of this term, is what the cube count uses. □

The full dictionary is Table 2 in Appendix A. It includes the all-integer square, the coefficient conjugation and the unit-coordinate support. The position cutoffs in LWW Sections 4.2–4.7 are sharp; η is a cutoff in the list product. There is no smooth cutoff in the second position coordinate.

Proposition 5.2 (Root averaging)**.** *Using the arguments in* H2′*, the block version of LWW Proposition 4.4 [6, pp. 22–24] is*

$$\begin{aligned} &\frac{1}{d_0 Y M N} \sum_{P \in \Omega \cap \mathcal{C}_{\mathcal{N}}} \langle u_P, (\mathcal{A}_{\mathcal{N}} \mathcal{A}_{\mathcal{N}}^*)^R u_P \rangle \\ &\quad = c_{\mathcal{N}} \int_1^{16} \int_1^2 \int_0^1 M_R^{(\mathcal{N})}(d_0 Y M s, N t_0, \beta)\, d\beta\, dt_0\, ds + O_A(L^{-2AR}). \end{aligned} \tag{5.4}$$

Here $M_R^{(\mathcal{N})}$ is LWW's M_R with g' replaced by $g_{\mathcal{N}}$ and LWW's coordinate-parity test replaced by the labelled state space of Section 3.4.

Proof. A closed path from a root $P \in \mathcal{C}_{\mathcal{N}}$ has $P_j = g_{\mathcal{N}}(P) z_j$, with z_j primitive, $z_0 = z_{2R} = (1,0)^T$ and $|z_j| \le 2^7 d_0 Y$. List divisibility becomes $[z_j]_p = \ell_p(P) = \ker(u, c') \bmod p$. Its good-state ratio is (3.17) at the current state. Apply the path expansion and cube argument of LWW Proposition 4.4 with Lemma 4.1 in place of its root count. The measure (4.29), p. 23, becomes $c_{\mathcal{N}}\, d(u/(d_0 Y M))\, d(v/N)\, d\beta$, and each prescribed line has mass $1/(p+1)$.

The absolute coefficient mass remains $\exp(O_K(L^{0.73}))$, since $|z_j| \ll R d_0 Y$. The Bonferroni truncation has $S \le \exp(O(L^{0.96})) \le \exp(C_0 L^{0.98})$. By (5.3) and

$$\partial_\beta \frac{\tau_{\mathcal{N}}(w)}{\tau_{\mathcal{N}}(z)} = \frac{\det(z,w)}{\tau_{\mathcal{N}}(z)^2} = \frac{1}{\tau_{\mathcal{N}}(z)^2} \quad (\det(z,w) = 1), \tag{5.5}$$

the boundary surfaces have a transverse derivative at least $1/16$ and gradients $x^{o(1)}$. The neighborhood volume is $\ll \varepsilon \max |\nabla E|/|\partial_{t_0} E| = x^{-c\delta/8 + o(1)}$ at the chosen cube size. Sum the per-cube power errors before the moment limit; their product with the coefficient mass gives $O_A(L^{-2AR})$. These are the changes to the stated LWW argument. □

Proposition 5.3 (Block Type II estimate)**.** *Assume* H2′*. Let* $M, N \geq x^{\delta}$*,* $MN \asymp x$ *and let the coefficients satisfy the sifting and bound hypotheses above. After choosing* A_0 *and then* K *sufficiently large, independently of the fixed band exponents* a_i*,*

$$|Q_Y^{\mathrm{all}} - Q_Y^{\mathrm{maj},\mathcal{N}}| \ll_A \mathcal{N}YL^{-A}. \tag{5.6}$$

Proof. We apply the arguments of LWW Sections 4.4–4.7 as written, with the replacements in Table 2. Section 4.4 uses relative coordinates, the linear form τ and the projection χ; use $\tau_{\mathcal{N}}$ and $\chi_{\mathcal{N}} = \mathbf{1}_{g_{\mathcal{N}}z\in\Omega}$. Its auxiliary operators have the same relative determinant, shared-prime lines and intrinsic tests. In (4.40)–(4.41), [6, pp. 27–28],

$$z' = j_1 w_z + hz, \qquad \frac{\tau_{\mathcal{N}}(z')}{\tau_{\mathcal{N}}(z)} = h + j_1 \frac{\tau_{\mathcal{N}}(w_z)}{\tau_{\mathcal{N}}(z)}. \tag{5.7}$$

The ratio strip leaves $O(1)$ integers h, uniformly in j_1. The clean and dirty bounds (4.42)–(4.50) use this geometry and the same bounded endpoint multipliers. The combinatorial signed-memory expansion of Section 4.5, pp. 32–35, uses these operator bounds, root averaging and bounded measure mass; $c_{\mathcal{N}} \leq 1$ suffices.

For the endpoint pairing in Section 4.6, use the inner product conjugate-linear in its initial entry and set

$$f(Dbm, s) = \overline{\alpha_m}\beta_s. \tag{5.8}$$

At $P = (Dbm, s)$, $Q = (Dar, n)$ this gives exactly

$$\overline{f(P)}f(Q) = \alpha_m\overline{\alpha_r}\beta_n\overline{\beta_s}. \tag{5.9}$$

All support, list-invariance and norm bounds are unchanged by this conjugation. Every prime divisor of Db is below Z, whereas s is Z-rough; a common prime of the endpoint coordinates must therefore divide m, s. If $p \mid (m, s)$ on endpoint support, then $p > Z$ and $p \nmid \mathcal{N}$; the determinant equation gives $p \mid b - a$. The retained cross-edge exclusion gives $b \neq a$, while $0 < |b - a| < 4Y < Z$, a contradiction. The argument is independent of the sign of the common cofactor.

Use Lemma 3.4 for the partition in Section 4.6, and the positive matrix and ambient spectral Hölder argument in Lemma 3.7 for (4.64)–(4.68), pp. 36–37. The trace loss is $(d_0Y)^{1/(2R)} = O(1)$. Section 4.7 has $O(1)$ exact-edge and $O(Y)$ comparison-edge counts in the reduced determinant, as in (4.70), p. 37. For (4.75), p. 38, fix $k_0 = mn$, $\ell_0 = rs$. Then

$$\ell_0 \equiv ba^{-1}k_0 \pmod{\mathcal{N}}, \qquad |bk_0 - a\ell_0| \ll \mathcal{N}Y. \tag{5.10}$$

The interval for ℓ_0 has length $O(\mathcal{N})$ and one residue class, so there are $O(1)$ choices. Raw absolute estimates use the all-integer cofactor range of Proposition 2.5; the extra $\mathcal{N}/X$ costs a fixed logarithmic power.

The moment/root normalization itself remains d_0YMN, giving the pairing scale XY of LWW. The right side $\mathcal{N}Y$ in (5.6) is a permitted relaxation by $\mathcal{N}/X \ll L^B$, absorbed by increasing the fixed precision. It is independent of the Cauchy cofactor factor $\mathcal{N}/Y$. □

Remark 5.4 (Determinant scale). Using arcs for the unreduced determinant without the block congruence would connect a position to $\asymp \mathcal{N}Y$ positions. The trace loss would be $(d_0Y\mathcal{N})^{1/(2R)} = \exp(L^{1/2} + o(L^{1/2}))$ rather than $O(1)$. The block congruence absorbs this determinant scale and reappears in the major arcs.

6 Major arcs

6.1 Reduction to the averaged energy

Proposition 6.1 (Major-arc reduction)**.** *For every fixed $D_* > 0$, (α^*) at a sufficiently large fixed precision $A = D_* + O(A_0 + B_1)$ implies*

$$|Q_Y^{\mathrm{maj},\mathcal{N}}| \ll_{D_*} \mathcal{N} Y L^{-D_*}. \tag{6.1}$$

Proof. By Lemma 3.1, on the six-variable unit support,

$$\begin{aligned} H_{\mathcal{N}}(bmn - ars; a, b) &= \psi_1(t'/Y)\mathbf{1}_{bmn\equiv ars\ (\mathcal{N})} \int_{\mathfrak{M}_1} e(\phi(t' - b + a))\, d\phi, \\ t' = (bmn - ars)/\mathcal{N}, &\qquad \mathbf{1}_{bmn\equiv ars\ (\mathcal{N})} = \frac{1}{\varphi(\mathcal{N})} \sum_{\chi \bmod \mathcal{N}} \chi(bmn)\overline{\chi(ars)}. \end{aligned} \tag{6.2}$$

Write $\phi = h_0/k + \lambda_0/Y$, $(h_0, k) = 1$, $k \le L^{A_0}$, $|\lambda_0| \le 2L^{A_0}$, and $d\phi = d\lambda_0/Y$. Its factor is

$$e(h_0 t'/k)e(-h_0(b-a)/k)e(\lambda_0 t'/Y)e(-\lambda_0(b-a)/Y). \tag{6.3}$$

The factors in a, b modulo k expand in characters, with mass at most k each, since a, b are units modulo k.

For $e(h_0 t'/k)$, put $u_1 = bmn$, $u_1' = ars$. Both are units modulo $k\mathcal{N}$, and

$$\mathbf{1}_{\mathcal{N}|u_1 - u_1'} e(h_0 t'/k) = \mathbf{1}_{\mathcal{N}|u_1 - u_1'} e(h_0 u_1/(k\mathcal{N}))e(-h_0 u_1'/(k\mathcal{N})).$$

Let

$$H = \ker((\mathbb{Z}/k\mathcal{N}\mathbb{Z})^\times \longrightarrow (\mathbb{Z}/\mathcal{N}\mathbb{Z})^\times), \qquad |H| = k \prod_{\substack{p\mid k \\ p\nmid \mathcal{N}}} (1 - 1/p) \le k. \tag{6.4}$$

The reduction is surjective. For each unit class $a' \bmod \mathcal{N}$, choose a unit representative $\widetilde{a}' \bmod k\mathcal{N}$; its fibre is $\widetilde{a}' H$. Expand $w \mapsto e(h_0 \widetilde{a}' w/(k\mathcal{N}))$ in characters $\eta \in \widehat{H}$, with coefficients $|c_{a',\eta}| \le 1$. Every such character extends to a character ψ_η of the finite abelian group $(\mathbb{Z}/k\mathcal{N}\mathbb{Z})^\times$, independently of a'. Hence on that fibre

$$e(h_0 u_1/(k\mathcal{N})) = \sum_{\eta \in \widehat{H}} c_{a',\eta} \psi_\eta(\widetilde{a}')^{-1} \psi_\eta(u_1). \tag{6.5}$$

The same representative works on the u_1' side. Cauchy gives

$$\sum_{a'} \left| \sum_{\eta} c_{a',\eta} \psi_\eta(\widetilde{a}')^{-1} X_{a',\eta} \right|^2 \le k \sum_{\eta} \sum_{a'} |X_{a',\eta}|^2. \tag{6.6}$$

Thus class-dependent coefficients disappear at fixed powers of k, including when $(k, \mathcal{N}) > 1$. Together with the a, b factors, the character mass is at most k^4 (LWW use k^6).

For the smooth factor put $w = u_1/(\mathcal{N}Y)$, $w' = u_1'/(\mathcal{N}Y)$ and $\kappa_X = MN/\mathcal{N}$. The dyadic variables satisfy

$$w, w' \in [\kappa_X, 16\kappa_X], \qquad \kappa_X \gg L^{-B}, \quad \kappa_X \le 1, \qquad \psi_1(t'/Y)e(\lambda_0 t'/Y) = \psi_1(w - w')e(\lambda_0(w - w')). \tag{6.7}$$

If $\kappa_X \le 1/4$, the difference has absolute value at most 4 and $\psi_1 = 1$. Otherwise it varies on a bounded interval. Set $z = \log(w'/\kappa_X)$ and $\zeta = \log(w/w')$. With a fixed smooth cutoff ϖ equal to 1 on $[1, 16]$, the kernel is

$$\varpi(e^z)\varpi(e^{z+\zeta})\psi_1(\kappa_X e^z(e^\zeta - 1))e(\lambda_0\kappa_X e^z(e^\zeta - 1)). \tag{6.8}$$

Its support lies in a fixed box of side $O(1)$. In the variables $w/\kappa_X, w'/\kappa_X$, derivatives are bounded by fixed powers of L^{A_0} because $\kappa_X \le 1$. Consequently for a fixed sufficiently large B_0 the transform obeys

$$F_{\lambda_0}(v, s) \ll_j (1 + |v|/L^{B_0})^{-j}(1 + |s|/L^{B_0})^{-j}. \tag{6.9}$$

Taking B_1 sufficiently large leaves frequencies $|t|, |s| \le L^{B_1}$. This is the Mellin separation of LWW Section 5, pp. 42–43, at scale 1 rather than X. The cutoff here does not force a narrow product difference; the arithmetic is in the character congruence.

With the normalized polynomials (2.6), each product sum has size $YMN \cdot BMN$ with its displayed normalization. The $1/Y$ arc measure therefore gives the scale

$$\frac{(YMN)^2}{Y}\frac{1}{\varphi(\mathcal{N})}\sum_\chi \int |BMN|^2 = Y(MN)^2\frac{1}{\varphi(\mathcal{N})}\sum_\chi \int |BMN|^2. \tag{6.10}$$

To obtain (6.1) it suffices that the energy be $\ll L^{-D_*}\mathcal{N}/(MN)^2$, which follows from $L^{-A}/\mathcal{N}$ because $MN \le \mathcal{N}$. Cauchy in t, χ, and the dyadic tail estimates of LWW Section 5, p. 43, leave (α^*) with $\xi = \chi\psi$ on m, n and $\xi_b = \chi\psi_2$ on b, where $\psi = \psi_\eta$, $\psi_2 = \psi\psi_b'$ and ψ_b' is modulo k. The primed product uses $-\lambda_0, \overline{\alpha}, \overline{\beta}$; the estimates below also hold after these conjugations and the corresponding sign change of the frequency twist. The arc count L^{2A_0}, character mass k^4, λ_0 integral L^{A_0} and s integral L^{B_1} are fixed logarithmic costs, absorbed by $A = D_* + O(A_0 + B_1)$.

For completeness, Parseval for a product polynomial gives

$$\frac{1}{\varphi(\mathcal{N})}\sum_\chi \left|\sum_u c_u\chi(u)\psi(u)u^{it}\right|^2 = \sum_{a'\in(\mathbb{Z}/\mathcal{N}\mathbb{Z})^\times}\left|\sum_{u\equiv a'\ (\mathcal{N})} c_u\psi(u)u^{it}\right|^2 \ll (YX/\mathcal{N}+1)\sum_u |c_u|^2 \ll \frac{L^{O(1)}}{\mathcal{N}}. \tag{6.11}$$

Here the product length is $\asymp XY$ and $|c_u| \ll L^C\tau_3(u)/(XY)$. For MN alone the squared coefficient norm is $L^{O(1)}/(MN) \le L^{O(1)+B}/\mathcal{N}$; this extra power is absorbed. □

6.2 Prime sums from a fixed zero-free region

Lemma 6.2 (Lemma P). *Let $2 \le q \le \mathcal{N}^3$, and let $\chi \bmod q$ be induced by a primitive χ^* of conductor $q^* > 1$. Fix positive $C_{\mathrm{zf},1}, C_{\mathrm{zf},2}$ and $C_3 \ge 1$ with $C_{\mathrm{zf},2} \ge C_3 + 1$. Suppose $L(s, \chi^*) \ne 0$ in (1.3)'s rectangle for these constants. For $|u| \le L^{C_3}$ and $\mathcal{N}^{1/6} \le y \le \mathcal{N}$,*

$$S(y) := \sum_{\ell\le y}\Lambda(\ell)\chi(\ell)\ell^{iu} \ll yL^3(L^{-C_{\mathrm{zf},1}/12} + L^{-C_{\mathrm{zf},2}}). \tag{6.12}$$

The implied constant can be absolute once $L \ge L_0(C_{\mathrm{zf},1}, C_{\mathrm{zf},2}, C_3)$.

Proof. The imprimitive difference is supported on prime powers at primes dividing q, and contributes at most $\omega_{\mathrm{all}}(q)\log y \le 5L^2$, absorbed since $y \ge \mathcal{N}^{1/6}$. For χ^* put

$$F(s) = -\frac{L'}{L}(s - iu, \chi^*), \quad c = 1 + 1/L, \quad T = L^{C_{\mathrm{zf},2}} - L^{C_3} - 1 \ge \tfrac{1}{2}L^{C_{\mathrm{zf},2}}. \tag{6.13}$$

Truncated Perron [1, Chapter 17] gives

$$\sum_{\ell\le y}\Lambda(\ell)\chi^*(\ell)\ell^{iu}=\frac{1}{2\pi i}\int_{c-iT}^{c+iT}F(s)\frac{y^s}{s}\,ds+O(y(\log y)^2/T+\log y). \tag{6.14}$$

The usual error is the sum of $\Lambda(\ell)(y/\ell)^c\min(1,(T|\log(y/\ell)|)^{-1})$. The choice $y\le\mathcal{N}$ ensures $y^c\le ey$ and $\sum\Lambda(\ell)\ell^{-c}\ll L$.

For $1/2\le\sigma\le 2$, the partial-fraction formula and local zero count [1, Chapter 16] read

$$\frac{L'}{L}(s,\chi^*)=\sum_{|\gamma-t|\le 1}\frac{1}{s-\rho_0}+O(\log(q^*(|t|+2))),\quad N(t+1,\chi^*)-N(t,\chi^*)\ll\log(q^*(|t|+2)). \tag{6.15}$$

Put $\sigma_0=1-C_{\mathrm{zf},1}\log L/(2L)$. For $\sigma\ge\sigma_0$ and $|t|+1\le L^{C_{\mathrm{zf},2}}$, all nearby zeros have real part at most $1-C_{\mathrm{zf},1}\log L/L$ and horizontal distance at least $C_{\mathrm{zf},1}\log L/(2L)$. Their number is $O(\log(q^*L^{C_{\mathrm{zf},2}}))=O(L)$ after a threshold depending on the fixed constants. Hence

$$|L'/L(s,\chi^*)|\ll L\frac{2L}{C_{\mathrm{zf},1}\log L}+L\ll L^2. \tag{6.16}$$

The shifted $F(s)$ has this bound on $\sigma\ge\sigma_0$, $|t|\le T$.

Move the contour to σ_0. There is no zero pole in the rectangle: a pole $s=\rho_0+iu$ would have $\Re\rho_0\ge\sigma_0$ and $|\Im\rho_0|\le T+|u|\le L^{C_{\mathrm{zf},2}}-1$, contradicting the region. Also $\sigma_0>1/2$, so $s=0$ is outside, and χ^* is nonprincipal. The vertical and horizontal integrals are respectively

$$\ll y^{\sigma_0}L^2\int_{-T}^{T}\frac{dt}{|\sigma_0+it|}\ll y^{\sigma_0}L^3,\qquad\ll(c-\sigma_0)y^cL^2/T\ll yL^2/T.$$

Since $L/6\le\log y\le L$, $y^{\sigma_0}\le yL^{-C_{\mathrm{zf},1}/12}$. Combining these estimates yields

$$S(y)\ll yL^{3-C_{\mathrm{zf},1}/12}+yL^{2-C_{\mathrm{zf},2}}+y(\log y)^2/T+\log y+L^2\ll yL^3(L^{-C_{\mathrm{zf},1}/12}+L^{-C_{\mathrm{zf},2}}). \tag{6.17}$$

□

Remark 6.3. The width $C\log L/L$ is what gives the arbitrary fixed logarithmic saving on $y\ge\mathcal{N}^{1/6}$. A width $c/\log q^*$ with $q^*\asymp\mathcal{N}$ gives only a factor of the size $e^{-c/6}$. Only the logarithmic derivative on the contour is used.

Corollary 6.4 (Pointwise prime-factor polynomial)**.** *Suppose $\xi=\chi\psi \bmod k\mathcal{N}$ is induced by χ^* of conductor $L^{A_1}<q^*\le k\mathcal{N}$ and Lemma 6.2 applies. If*

$$C_{\mathrm{zf},1}\ge 12(D+4),\quad C_{\mathrm{zf},2}\ge D+4,\quad C_{\mathrm{zf},2}\ge C_3+1,\quad C_3=\max(B_1,2B)+1, \tag{6.18}$$

then, uniformly for $|t|\le T_0+1$ and sufficiently large L depending on the fixed parameters,

$$|N_\xi(t)|\le 8L^{1-D}+\frac{4Z+2L}{V}\le 9L^{1-D}. \tag{6.19}$$

Proof. Put $u=t+v$, so $|u|\le L^{C_3}$. Both R_Z and the unit indicator are completely multiplicative. Thus

$$\begin{aligned}c_V(n)R_Z(n)\mathbf{1}_{(n,\mathcal{N})=1}&=\sum_{\substack{\ell k'=n\\ \ell>V}}\Lambda(\ell)R_Z(\ell)\mathbf{1}_{(\ell,\mathcal{N})=1}R_Z(k')\mathbf{1}_{(k',\mathcal{N})=1},\\ N_\xi(t)&=\frac{1}{N}\sum_{k'<2N/V}R_Z(k')\mathbf{1}_{(k',\mathcal{N})=1}\xi(k')(k')^{iu}\sum_{\ell\in I_{k'}}\Lambda(\ell)R_Z(\ell)\mathbf{1}_{(\ell,\mathcal{N})=1}\xi(\ell)\ell^{iu},\\ I_{k'}&=(\max(V,N/k'),2N/k'].\end{aligned} \tag{6.20}$$

In a ratio-two interval at most one power of each prime is present. Dropping the indicators in the inner sum therefore costs at most $\sum_{p\le Z}\log p+\sum_{p\mid\mathcal N}\log p\le 2Z+L$, and after the k' sum at most $(4Z+2L)/V$. For every nonempty $I_{k'}$, the two endpoints are between V and $2N\le\mathcal N$, and

$$\log V=(\log x)/5\ge(L-B\log L-\log 2)/5\ge L/6$$

for sufficiently large L. The modulus is at most $\mathcal N^2$. Lemma 6.2 and (6.18) give $|S(y)|\le 2C_PyL^{-D-1}\le yL^{-D}$ for $L\ge 2C_P$. Consequently

$$|N_\xi(t)|\le 4L^{-D}(\log(2N/V)+1)+(4Z+2L)/V\le 8L^{1-D}+(4Z+2L)/V.$$

Since $V\ge\mathcal N^{1/6}$ and $Z=\exp(L^{0.24})$, the last ratio is at most L^{1-D} for large L. Only endpoint sums with $y\ge V$ are used, even when $I_{k'}$ is short. □

The mean value theorem used below has the form

$$\int_{-T}^{T}\left|\sum_u b_u u^{it}\right|^2dt=\sum_u|b_u|^2\bigl(2T+O(\delta_u^{-1})\bigr),\tag{6.21}$$

where δ_u is the separation of $\log u$ from the other logarithmic frequencies [8].

6.3 Exceptional values of the short polynomial

For $0\le i<2T_0$ let $J_i=[-T_0+i,-T_0+i+1]$. For each $\chi \bmod\mathcal N$ choose a point $t_{\chi,i}$ maximizing $|B_{\xi_b}M_\xi N_\xi|$ on J_i. Let $\mathcal P_0$ use the even i and $\mathcal P_1$ the odd i. Within each family, points for a fixed character are 1-spaced, and

$$\sum_\chi\int_{|t|\le T_0}|BMN|^2\,dt\le\sum_{\varepsilon=0,1}\sum_{(\chi,t)\in\mathcal P_\varepsilon}|B_{\xi_b}(t)M_\xi(t)N_\xi(t)|^2.\tag{6.22}$$

Below $\mathcal P$ is either family, and each argument applies to both.

Lemma 6.5 (Sobolev estimate)**.** *For $F\in C^1([-T_0-1,T_0+1])$ and 1-spaced points $t_1<\cdots<t_r$ in $[-T_0,T_0]$,*

$$\sum_j|F(t_j)|^2\le\int_{|t|\le T_0+1}|F|^2\,dt+2\left(\int_{|t|\le T_0+1}|F|^2\,dt\right)^{1/2}\left(\int_{|t|\le T_0+1}|F'|^2\,dt\right)^{1/2}.\tag{6.23}$$

Proof. Average $|F(t_j)|^2=|F(s)|^2+\int_s^{t_j}(|F|^2)'$ over $s\in[t_j-1/2,t_j+1/2]$. These intervals have disjoint interiors; use $|(|F|^2)'|\le 2|F||F'|$ and Cauchy–Schwarz. □

Lemma 6.6 (Mean value over residue classes)**.** *If (c_u) is supported on positive $u\le 4\mathcal N$ and $|\psi_3|\le 1$, then for $T\ge 1$,*

$$\sum_{\chi \bmod\mathcal N}\int_{-T}^{T}\left|\sum_u c_u\chi(u)\psi_3(u)u^{it}\right|^2dt\le\varphi(\mathcal N)(2T+C_0')\sum_u|c_u|^2\tag{6.24}$$

with absolute C_0'.

Proof. Character orthogonality gives

$$\sum_{\chi}\left|\sum_{u} c_u\chi(u)d_u\right|^2=\varphi(\mathcal{N})\sum_{a'\in(\mathbb{Z}/\mathcal{N}\mathbb{Z})^\times}\left|\sum_{u\equiv a'\ (\mathcal{N})} c_u d_u\right|^2.$$

Each class has at most four positive integers at most $4\mathcal{N}$. If $u<u'$ are in it, $u'\ge u+\mathcal{N}\ge (4/3)u$ since $u\le 3\mathcal{N}$. The Montgomery–Vaughan mean value theorem [8] therefore gives $C_0'=O(1/\log(4/3))$. For length at most $\mathcal{N}$, a unit class has at most one term and the integral in that class is exactly $2T$ times its squared coefficient. □

Lemma 6.7 (Lemma LS)**.** *Fix $A_s\ge 1$ and put $E=\{(\chi,t)\in\mathcal{P}:|B_{\xi_b}(t)|>L^{-A_s}\}$. For $L\ge L_0(K,a_K,A_s,B_1,C_B)$,*

$$|E|\le \exp\bigl(L^{1-a_K}(2A_s+K+10)\log L\bigr)\le \exp(L^{0.9}). \tag{6.25}$$

Proof. Let $j=\lfloor L/\log(16Y)\rfloor\ge 1$; then $(16Y)^j\le\mathcal{N}<(16Y)^{j+1}$. Write $B_{\xi_b}(t)^j=\sum_u c_u^{(j)}\xi_b(u)u^{it}$, where

$$c_u^{(j)}=\sum_{\substack{b_1\cdots b_j=u\\ b_i\in\mathcal{A}}}\prod_i c_{b_i},\qquad c_b=\frac{\eta(b/Y)e(\lambda_0 b/Y)}{Y\prod_i V_i},\qquad |c_b|\le C_B/(16Y). \tag{6.26}$$

This polynomial is supported on $[Y^j,(16Y)^j]\subset[1,\mathcal{N}]$. For each disjoint band the j prime factors of u, with multiplicity, can be assigned to the j slots in at most $j!$ ways. Hence there are at most $(j!)^K$ representations and

$$|c_u^{(j)}|^2\le (j!)^K\sum_{\text{reps}}\prod_i|c_{b_i}|^2,\qquad \sum_u|c_u^{(j)}|^2\le (j!)^K\left(\sum_b|c_b|^2\right)^j\le (j!)^K\left(\frac{C_B^2}{16Y}\right)^j. \tag{6.27}$$

Use Lemmas 6.5 and 6.6 on B^j; its derivative inserts a coefficient $i\log u$, with $|\log u|\le L$. Since $\xi_b=\chi\psi_2$ and $|\psi_2|\le 1$,

$$\sum_{(\chi,t)\in\mathcal{P}}|B_{\xi_b}(t)|^{2j}\le 3L\varphi(\mathcal{N})(2T_0+2+C_0')(j!)^K(C_B^2/(16Y))^j. \tag{6.28}$$

The left side is at least $|E|L^{-2jA_s}$. Since $Y^{-j}<16^{j+1}Y/\mathcal{N}$ and $\varphi(\mathcal{N})\le\mathcal{N}$,

$$|E|\le L^{2jA_s}3L(2T_0+2+C_0')(j!)^K C_B^{2j}16Y. \tag{6.29}$$

Every $b\in\mathcal{A}$ exceeds $\exp(L^{a_K})$, so $j\le L^{1-a_K}$, $\log j\le\log L$, and $\log Y\le 2\sum_i L^{a_i}+\log 16\le 3KL^{a_K}$. It follows that

$$\log|E|\le 2A_s j\log L+Kj\log j+2j\log C_B+\log Y+O(\log L)\le L^{1-a_K}(2A_s+K+10)\log L.$$

For fixed $a_K>1/10$, this is eventually at most $L^{0.9}/2$. □

Corollary 6.8 (Bulk estimate)**.** *For each family $\mathcal{P}$,*

$$\begin{aligned}\sum_{\mathcal{P}\setminus E}|BMN|^2&\le L^{-2A_s}\sum_{\mathcal{P}}|M_\xi N_\xi|^2\\ &\le L^{-2A_s}3L\varphi(\mathcal{N})(2T_0+O(1))\frac{L^2\sum_{u\le 4MN}\tau(u)^2}{(MN)^2}\\ &\ll L^{-2A_s+B_1+B+8}\frac{\varphi(\mathcal{N})}{\mathcal{N}}.\end{aligned} \tag{6.30}$$

Proof. The product MN polynomial is supported in $(MN,4MN]$, with $4MN\le 8x\le 4\mathcal{N}$ and coefficients at most $L\tau(u)/(MN)$. Apply Lemmas 6.5 and 6.6, and $\sum_{u\le y}\tau(u)^2\ll y(\log y)^3$. □

6.4 Energy on the exceptional set

Lemma 6.9 (Lemma HM)**.** *Let $(\xi_r, t_r)_{r\le R'}$ be character-frequency pairs modulo q, 1-spaced in t whenever their characters agree. Let a_m be supported on $(X', X'']$, $X' < X'' \le 8X'$, and let $w \ge 0$ be at least 1 on that interval. Then*

$$\sum_r \left|\sum_m a_m \xi_r(m) m^{it_r}\right|^2 \le \|a\|_2^2 \max_r \sum_s |G_{rs}|, \qquad G_{rs} = \sum_m w(m)\xi_r(m)\overline{\xi_s(m)} m^{i(t_r - t_s)}. \tag{6.31}$$

Proof. Put $S_r = \sum_m a_m \xi_r(m) m^{it_r}$ and $b_r = \overline{S_r}/(\sum_s |S_s|^2)^{1/2}$; the zero case is immediate. Cauchy gives

$$(\sum_r |S_r|^2)^{1/2} = \sum_r b_r S_r \le \|a\|_2 \left(\sum_m w(m)\left|\sum_r b_r \xi_r(m) m^{it_r}\right|^2\right)^{1/2}.$$

The squared last factor is $\sum_{r,s} b_r \bar{b}_s G_{rs}$. The Hermitian Schur bound and $\sum |b_r|^2 = 1$ prove the claim. □

Choose $q = k\mathcal{N}$ and a smooth w_0 with $\mathbf{1}_{(X',X'']} \le w_0 \le \mathbf{1}_{[X'/2,16X']}$ and $|w_0^{(i)}| \le C_i (X')^{-i}$. There are three kinds of Gram entry.

1. On the diagonal, $G_{rr} \le 16X'$.

2. If $\xi_r = \xi_s$, $r \ne s$, and $\tau = t_r - t_s$, then $|\tau| \ge 1$ and

$$G_{rs} = \sum_{d|q} \mu(d) d^{i\tau} \sum_{m'} w_0(dm')(m')^{i\tau}. \tag{6.32}$$

   For $d \le 16X'$, Euler summation replaces the inner sum by its integral with error $O(1 + |\tau|)$: for $f(y) = w_0(dy)y^{i\tau}$,

$$\int |f'| \le \int |w_0'| + |\tau| \int_{X'/(2d)}^{16X'/d} dy/y \le C(1 + |\tau|).$$

   Two integrations by parts give integral size $O((X'/d)(1 + |\tau|)^{-2})$. For $d > 16X'$ the sum is empty. Thus

$$|G_{rs}| \le C' X' (1 + |\tau|)^{-2} \log L + \tau(k\mathcal{N})(1 + |\tau|), \tag{6.33}$$

   using $\sum_{d|q} 1/d \ll \log\log q$. The 1-spacing gives

$$\sum_{\substack{s\ne r \\ \xi_s = \xi_r}} |G_{rs}| \le C'' X' \log L + (2T_0 + 1)^2 \tau(k\mathcal{N}). \tag{6.34}$$

3. If $\xi_r \ne \xi_s$, their quotient is nonprincipal. On units it is $\chi_r \overline{\chi_s}$, and these are distinct characters modulo $\mathcal{N}$. Partial summation gives

$$|G_{rs}| \le C(1 + |\tau|)\Delta_\xi(16X'), \qquad \Delta_\xi(Y') = \max_{y \le Y'} \left|\sum_{m \le y} \xi(m)\right|, \quad |\tau| \le 2T_0 + 1. \tag{6.35}$$

If ξ is induced by primitive $\xi^* \bmod q^* > 1$, put $q_1 = \prod_{p \mid kN, p \nmid q^*} p$. Then

$$\sum_{m \le y} \xi(m) = \sum_{d \mid q_1} \mu(d)\xi^*(d) \sum_{m' \le y/d} \xi^*(m'), \qquad \Delta_\xi(Y') \le 2^{\omega_{\mathrm{all}}(kN)} \Delta_{\xi^*}(Y'). \tag{6.36}$$

The two inputs are Pólya–Vinogradov and Burgess at $r = 2$:

$$\Delta_{\xi^*}(Y') \le C(q^*)^{1/2} \log q^* \le C(L^{A_0}N)^{1/2}\, 2L, \tag{6.37}$$

$$\left| \sum_{H_0 < n \le H_0 + H} \xi^*(n) \right| \ll_\varepsilon H^{1/2}(q^*)^{3/16+\varepsilon} \quad (H \ge 1), \tag{6.38}$$

where (6.38) holds for every modulus, as in [4, introduction, (1)] (arbitrary moduli are allowed for $r \le 3$). Also

$$2^{\omega_{\mathrm{all}}(kN)} \le 2^{A_0 \log L} 2^{\omega_{\mathrm{all}}(N)} \le L^{A_0} \exp(O(L/\log L)) = N^{o(1)}, \qquad \tau(kN) = N^{o(1)}.$$

Corollary 6.10 (Exceptional energy)**.** *Let E be as in Lemma 6.7 and fix $\eta_0 = 1/40$. Then*

$$M \ge N^{3/8+\eta_0} \implies \sum_E |M_\xi(t)|^2 \ll \log L, \tag{6.39}$$

$$\sum_E |M_\xi(t)|^6 \ll L^9, \tag{6.40}$$

$$N \ge N^{1/2+2\eta_0} \implies \sum_E |N_\xi(t)|^2 \ll L^2 \log L. \tag{6.41}$$

Proof. For (6.39) use $a_m = \alpha_m/M$, $\|a\|_2^2 \le 2/M$, $X' = M$ and Burgess with $\varepsilon = \eta_0/8$. The distinct-character row contribution is at most

$$\begin{aligned} &|E| C(2T_0+2) 2^{\omega_{\mathrm{all}}(kN)} C_\varepsilon (16M)^{1/2} (L^{A_0}N)^{3/16+\varepsilon} \\ &\quad \le M N^{-\eta_0/2+\varepsilon+o(1)} \exp(L^{0.9}) = M N^{-3/320+o(1)} \exp(L^{0.9}) = o(M). \end{aligned} \tag{6.42}$$

Together with $16M + C''M \log L + (2T_0+1)^2 \tau(kN)$ this proves (6.39). For (6.40), $M^3 \ge (x^{1/5}/2)^3 \ge N^{3/5-o(1)}$. The coefficients of M_ξ^3 are at most $M^{-3}\tau_3(u)$ on $(M^3, 8M^3]$, with squared norm $\ll L^8/M^3$. Pólya–Vinogradov makes the distinct-character row at most

$$|E| L^{O(1)} N^{1/2+o(1)} \le M^3 N^{1/2-3/5+o(1)} \exp(L^{0.9}) = o(M^3),$$

yielding L^9. For (6.41), take $a_n = \beta_n/N$, $\|a\|_2^2 \le 2L^2/N$; Pólya–Vinogradov gives row $\le N N^{-2\eta_0+o(1)} \exp(L^{0.9}) = o(N)$. In all cases the same-character row is $O(X' \log L)$, because $X' \ge N^{1/5-o(1)}$ absorbs $(2T_0+1)^2 \tau(kN)$. □

Input 6.11 (Twisted sifted Möbius sums)**.** *LWW Proposition 2.2, (2.5), p. 5, states that for fixed $\delta, B', C' > 0$, $x^\delta \le D_1 \le 2x$, $I \subset [D_1, 2D_1]$ and a character of modulus $q \le (\log x)^{B'}$, for every fixed $A' > 0$,*

$$\sup_{|t| \le x^{C'}} \left| \sum_{d \in I} \mu(d) R_Z(d) \chi(d) d^{it} \right| \ll_{A', B', C', \delta} D_1 (\log x)^{-A'}. \tag{6.43}$$

For a bounded-variation weight w, (2.11), p. 6, multiplies this bound by $\|w\|_\infty + \mathrm{Var}_I(w)$. Fixed logarithmic powers in these norms are absorbed by a larger fixed precision.

6.5 The energy theorem

Theorem 6.12 (Theorem A)**.** *Fix A and $A_1 \geq 1$, and put*

$$\begin{aligned} D &= \left\lceil \frac{3}{2}(A + B_1 + B + 20) \right\rceil, \qquad C_3 = \max(B_1, 2B) + 1, \\ C_{\mathrm{zf},2} &= \max(D + 4, C_3 + 1), \qquad C_{\mathrm{zf},1} = 12(D + 4), \\ A_s &= A + B_1 + B + 10. \end{aligned} \tag{6.44}$$

In the setting of Section 6.1, $\mathrm{H1}^\star(C_{\mathrm{zf},1}, C_{\mathrm{zf},2}; A_1)$ *at* $\mathcal{N}$ *implies* (α^*)*, for a threshold depending on all fixed parameters and* A_1*.*

Proof. Split the characters into $\mathcal{X}_s = \{\chi : \mathrm{cond}(\chi\psi) \leq L^{A_1}\}$ and its complement $\mathcal{X}_b$. For $\chi \in \mathcal{X}_s$, let ξ^* induce $\xi = \chi\psi$. On the coefficient support,

$$M_\xi(t) = \frac{1}{M} \sum_m \mu(m) R_Z(m) \mathbf{1}_{(m,\mathcal{N})=1} \xi^*(m) m^{i(t+v)}. \tag{6.45}$$

Removing the unit indicator changes this by at most

$$\sum_{\substack{p \mid \mathcal{N} \\ p > Z}} \frac{2M/p}{M} \leq 2\omega_{\mathrm{all}}(\mathcal{N})/Z \leq 2Le^{-L^{0.24}}.$$

Apply LWW Proposition 2.2, (2.5), p. 5, and (2.11), p. 6, to the sum without the indicator: the modulus is at most L^{A_1}, $M \geq x^\delta$ and $|t + v| \leq x$ for large x. For every fixed A' this gives $\sup_{|t| \leq T_0} |M_\xi(t)| \ll L^{-A'}$. This is the use of the sifted Möbius input, whose proof cites [7, Lemma 3.9(ii)]. For fixed ψ, the primitive inducing character determines χ, so $|\mathcal{X}_s| \leq L^{2A_1}$. The polynomial $B_{\xi_b} N_\xi$ is supported on $u \leq 32YN$ with coefficients at most $C_B L\tau(u)/(16YN)$ and squared coefficient norm $\ll L^5/(YN)$. The ordinary mean value theorem gives $\int_{|t| \leq T_0} |BN|^2 \ll (T_0 + YN)L^5/(YN) \ll L^5$. Thus with $A' = A + A_1 + 4$,

$$\frac{1}{\varphi(\mathcal{N})} \sum_{\chi \in \mathcal{X}_s} \int |BMN|^2 \ll \frac{L^{2A_1 - 2A' + 5}}{\varphi(\mathcal{N})} \ll L^{-A}/\mathcal{N}, \tag{6.46}$$

using $\mathcal{N}/\varphi(\mathcal{N}) \ll \log L$.

For $\chi \in \mathcal{X}_b$, $q^* \mid k\mathcal{N}$ and $L^{A_1} < q^* \leq k\mathcal{N}$. The assumed region and Corollary 6.4 yield $|N_\xi(t)| \leq 9L^{1-D}$ for $|t| \leq T_0 + 1$. Discretize on this subset as above; restricting characters only reduces the bounds for $\mathcal{P}$ and E. The bulk is at most

$$L^{-2A_s + B_1 + B + 8} \varphi(\mathcal{N})/\mathcal{N} \leq L^{-A-1} \varphi(\mathcal{N})/\mathcal{N}.$$

If $M \geq \mathcal{N}^{3/8 + \eta_0}$, (6.39) gives

$$\sum_E |BMN|^2 \leq C_B^2 \sup_E |N|^2 \sum_E |M|^2 \ll L^{2-2D} \log L. \tag{6.47}$$

If $M < \mathcal{N}^{3/8 + \eta_0}$, then

$$\begin{aligned} N = \frac{MN}{M} \gg \frac{x}{M} \gg \frac{\mathcal{N}^{5/8 - \eta_0}}{L^B}, \\ N \geq \mathcal{N}^{1/2 + 2\eta_0} \quad \text{for sufficiently large } \mathcal{N}. \end{aligned} \tag{6.48}$$

Hölder with exponents $3, 3/2$, followed by (6.40) and (6.41), gives

$$\begin{aligned}\sum_E |B|^2|M|^2|N|^2 &\le C_B^2 \sup_E |N|^{2/3} \sum_E |M|^2|N|^{4/3} \\ &\le C_B^2 (9L^{1-D})^{2/3} (\sum_E |M|^6)^{1/3} (\sum_E |N|^2)^{2/3} \ll L^{(2/3)(1-D)+3+2}.\end{aligned} \tag{6.49}$$

In either case the exceptional contribution is

$$\ll L^{6-2D/3} \le L^{-A-B_1-B-14} \le L^{-A-1}. \tag{6.50}$$

One may also use the looser exponent $(2/3)(1-D)+6$ in the second case and $3-2D$ with a factor $\log L$ in the initial case; both leave the same required margin. Sum the two families and divide by $\varphi(\mathcal{N})$:

$$\frac{1}{\varphi(\mathcal{N})} \sum_{\chi \in \mathcal{X}_b} \int |BMN|^2 \ll L^{-A-1}/\mathcal{N} + L^{-A-1}/\varphi(\mathcal{N}) \ll L^{-A}/\mathcal{N}.$$

□

The inputs to Theorem 6.12 are the ordinary mean value theorem, Pólya–Vinogradov, Burgess at $r = 2$, partial fractions and zero counting, Perron, and LWW Proposition 2.2. The half-plane (1.2) enters only by supplying (1.3). For one fixed region the theorem supplies its selected fixed precision A; it does not assert all precisions from one fixed width.

6.6 The small-radical input

Input 6.13 (Iwaniec, 1974, Theorem 2)**.** *For $q \ge 3$, put $d = \operatorname{rad}(q)$, $\ell = \log(q(|t|+3))$ and*

$$\vartheta^{-1} = 4 \cdot 10^4 \{\log d + (\ell \log(2\ell))^{3/4}\}. \tag{6.51}$$

There is at most one nonprincipal character modulo q and one zero ρ_0 in $\Re\rho_0 > 1-\vartheta$; if present, the character is real and the zero real and simple [5, Theorem 2, p. 97].

Lemma 6.14 (Lemma Q)**.** *If $\operatorname{rad}(\mathcal{N}) \le L^C$ and $k \le L^{A_0}$, every real primitive character of conductor $q^* \mid k\mathcal{N}$ has*

$$q^* \le 8 \operatorname{rad}(k\mathcal{N}) \le 8L^{A_0+C} \le L^{A_1}, \qquad A_1 = A_0 + C + 1, \quad L \ge 8. \tag{6.52}$$

Proof. A real primitive character is a Kronecker symbol with fundamental discriminant of absolute value m, $4m$ or $8m$, with m odd and squarefree [1, Chapter 5]. Hence $q^* \mid 8\operatorname{rad}(q^*) \mid 8\operatorname{rad}(k\mathcal{N})$. □

Corollary 6.15 (Small-radical region)**.** *For $\operatorname{rad}(\mathcal{N}) \le L^C$ and $A_1 = A_0 + C + 1$, (1.3) holds for every fixed $C_{\mathrm{zf},1}, C_{\mathrm{zf},2}$ once $L \ge L_0(C_{\mathrm{zf},1}, C_{\mathrm{zf},2}, C, A_0)$.*

Proof. For $q^* \mid k\mathcal{N}$, $q^* > L^{A_1}$, apply Input 6.13 with $q = q^*$. Here $\operatorname{rad}(q^*) \le L^{A_0+C}$ and $\ell = \log(q^*(|t|+3)) \le 2L$ at the fixed height $L^{C_{\mathrm{zf},2}}$. Thus

$$\begin{aligned}\vartheta^{-1} &\le 4 \cdot 10^4 \{(A_0+C)\log L + (2L\log(4L))^{3/4}\}, \\ \vartheta &\ge c_I L^{-3/4} (\log L)^{-3/4} > C_{\mathrm{zf},1} \log L / L\end{aligned} \tag{6.53}$$

for sufficiently large L, with absolute $c_I > 0$ of size 10^{-5}. A real exceptional character would contradict Lemma 6.14, so the exceptional case is absent. The strict width comparison also gives the closed boundary in (1.3). □

The explicit radical dependence is needed because k contributes radical up to L^{A_0} even when $\mathcal{N} = 2^m$; a prime-power region alone does not cover these conductors.

Corollary 6.16 (The analytic alternatives)**.** *Theorem 6.12 supplies* (α^*) *from the selected* H1**, and therefore from* H1 *for every even* $\mathcal{N}$*. In the class* $\mathrm{rad}(\mathcal{N}) \le L^C$ *it follows from Corollary 6.15. For* $0 < |h| \le L^C$*, all characters have conductor at most* L^{A_0+C}*, but the scale-*1 *Mellin reduction does not apply; Section 11.4 uses LWW Section 5 instead.*

6.7 The classical-region obstruction

Remark 6.17 (A separate contour estimate). Fix $C_t \ge 2$. Under (1.2), formula (5.15) of LWW, p. 44, separates a prime in $\mathcal{P}_1$ from B_{ξ_b}. Lemma 3.5 of [2], followed by dyadic summation and partial summation, gives

$$\sum_{p\in\mathcal{P}_1} \frac{\xi_b(p)p^{it'}}{p} \ll L^{C_{\mathrm{prime}}+1}\exp(-L^{a_1}/16) \tag{6.54}$$

uniformly for $|t'| \le \mathcal{N}^{C_t}$ and nonprincipal ξ_b of modulus $k\mathcal{N} \le \mathcal{N}^2$. Here C_{prime} is fixed, depending on C_t and the fixed exponent ranges. The cited Perron argument uses $\Re s = 15/16$ and $L'/L \ll \log^2(q(|t|+3))$.

For the classical real-zero lower bound $1-\beta_1 \gg_\varepsilon q^{-\varepsilon}$, even at $q \le \exp(L^{a_1/2})$ the guaranteed lower bound for $(1-\beta_1)L^{a_1}$ can tend to zero. It gives no divergence to infinity, and leaves a possible exceptional conductor in $(L^C, \exp(L^{a_1/2})]$ outside this argument.

Proposition 6.18 (Obstruction (EC))**.** *Let* $\chi \bmod q$ *be real primitive, with* $q \mid \mathcal{N}$*. For* $p \nmid \mathcal{N}$ *one has*

$$\mu(\mathcal{N}-p) = \chi(-1)\chi(p)\varepsilon(\mathcal{N}-p)\mu^2(\mathcal{N}-p), \qquad \varepsilon(n) = \prod_{p'\mid n}(-\chi(p')). \tag{6.55}$$

Suppose every split prime ($\chi(p')=1$) *at most* $\log q$ *divides* $\mathcal{N}$*, and at most* $\pi(\mathcal{N})/20$ *primes* $p < \mathcal{N}$ *have either* $\chi(p) = 1$ *or a split prime factor exceeding* $\log q$ *in* $\mathcal{N}-p$*. Then, for sufficiently large such* $\mathcal{N}$*,*

$$\left|\sum_{p<\mathcal{N}} \mu(\mathcal{N}-p)\right| \ge 0.27\pi(\mathcal{N}). \tag{6.56}$$

Proof. For squarefree n coprime to q, $\varepsilon(n) = \mu(n)\chi(n)$; for nonsquarefree n both sides of (6.55) vanish after multiplication by $\mu^2(n)$. Also $\chi(\mathcal{N}-p) = \chi(-1)\chi(p)$ when $p \nmid \mathcal{N}$, proving the identity. Outside the stated exceptional set, $\chi(p) = -1$. No split prime can divide $\mathcal{N}-p$: a small split prime divides $\mathcal{N}$ and would then divide p, contrary to $p \nmid \mathcal{N}$; a larger split prime is excluded by hypothesis. Thus $\varepsilon(\mathcal{N}-p) = 1$, and each squarefree such value has $\mu(\mathcal{N}-p) = -\chi(-1)$. Section 9, whose proof uses only Siegel–Walfisz, gives $(A_{\mathcal{N}}+o(1))\pi(\mathcal{N})$ squarefree values uniformly. Removing the exceptional set loses at most $\pi(\mathcal{N})/20$ of them, and their possible opposite contribution loses another $\pi(\mathcal{N})/20$. The $O(\omega_{\mathrm{all}}(\mathcal{N}))$ nonunit primes are negligible. Hence the left side of (6.56) is at least $(0.3739 - 2/20 + o(1))\pi(\mathcal{N}) > 0.27\pi(\mathcal{N})$. □

This proposition establishes an obstruction under its split-prime hypotheses. The classical lower bound $1-\beta \gg_\varepsilon q^{-\varepsilon}$ does not by itself exclude that scenario for a large conductor dividing $\mathcal{N}$. The possible zero-size condition $(1-\beta)\log\mathcal{N} \to 0$ does not replace the separate split-prime hypotheses in this proposition. It identifies why the present uniform argument uses (1.3). In the small-radical class all the real conductors dividing $k\mathcal{N}$ are small by Lemma 6.14.

7 Type I and the common divisor

Proposition 7.1 (Type I for the Goldbach shift)**.** *For the bounded function in Proposition 2.3, LWW Proposition 3.1 with the following changes gives* $T_1(F), T_2(F) \ll_A xL^{-A}$*, uniformly in the range* (2.1).

Proof. Use the argument of LWW Section 3, pp. 10–14. The classes with shift +2 become, for example, $k \equiv \mathcal{N}\bar{q} \pmod{d}$; every divisor d built from list primes is coprime to $\mathcal{N}$. The sifted modulus q can have a prime $> Z$ in common with $\mathcal{N}$. Discarding those terms costs

$$\ll xL^3\omega_{\mathrm{all}}(\mathcal{N})/Z = O_A(xL^{-A}). \tag{7.1}$$

At this point the unit condition on the sifted inner variable can also be removed at that cost. It is removed before expanding R_Z, so that no additional condition on the later Möbius variable is imposed.

After the truncated inclusion–exclusion in LWW Section 3, p. 11, r is odd and squarefree with small prime factors, and may share primes with $\mathcal{N}$. Put $g = (r, \mathcal{N})$. The list divisor d is coprime to $r\mathcal{N}$, so dg is squarefree. Use the exact identity (3.4), p. 12, in the form

$$\mu(dgv) = \mu(dg) \sum_{\substack{a \mid v \\ p \mid a \Rightarrow p \mid dg}} \mu(v/a). \tag{7.2}$$

The substitution is now

$$\mathcal{N} - qrj = dgat, \qquad t \equiv (\mathcal{N}/g)(da)^{-1} \pmod{qr/g}, \qquad t \geq 1. \tag{7.3}$$

The inverse exists: primes of a lie in dg, so $(da, qr/g) = 1$. Indeed d is disjoint from qr and the primes of g have been removed from the squarefree r; q is Z-rough. Also $(\mathcal{N}/g, r/g) = 1$, $(\mathcal{N}, q) = 1$ and $(g, qr/g) = 1$. Thus the class is reduced. Primes of a may divide g; no condition $(g, da) = 1$ is needed or imposed. Since qr, dga are odd and $\mathcal{N}$ is even, j odd is equivalent to t odd. The Möbius variable therefore lies in a reduced class modulo $2qr/g$, with 0 excluded.

Apply Bombieri–Vinogradov in LWW (3.1), p. 10, with the τ-weighted grouping of p. 13. For each fixed g, qr/g determines qr and its rough part q and small-prime part r uniquely; one may bound coefficients by L or by $L\tau(qr/g)$ without a $\tau(g)$ loss. The level is comparable to $\mathcal{N}/(dga)$ and depends on g. Apply the bound separately for each g and then sum its reciprocal weight:

$$\sum_{g \mid \mathrm{rad}(\mathcal{N})} \frac{1}{g} = \prod_{p \mid \mathcal{N}} (1 + 1/p) \ll \log L. \tag{7.4}$$

The sum over a of $1/a$ has the same admissible Euler-product bound as in LWW. For the large-a part in (3.5), p. 13, the new Rankin factor is

$$\prod_{p \mid g} (1 - p^{-1/20})^{-1} \leq 19^{R''} = L^{O(1)}, \qquad R'' = \lceil C' \log L \rceil, \tag{7.5}$$

absorbed by $x^{-1/50}$. Here r has at most R'' prime factors. The incompatibility $(r, d) > 1$ is unchanged: a list prime in both would divide $\mathcal{N}$.

The positive cofactor interval is

$$\left[\frac{\mathcal{N} - 2x}{dga}, \frac{\mathcal{N} - x}{dga} \right), \qquad \mathcal{N}/(dga) \ll L^B x/(dga). \tag{7.6}$$

The maximum over endpoints in Bombieri–Vinogradov handles this interval. Replacing x by $\mathcal{N}$ in the level costs only the fixed power L^B, covered by the chosen precision. Partial summation for T_1 is that of LWW Section 3, p. 14. □

8 Removal of the auxiliary weights

Proposition 8.1 (Normalized weight removal)**.** *For each fixed b choose b disjoint copies W_j of (2.3), and let m_j be their reduced-class means for $W_j(\mathcal{N}-a)$. Then $m_j = 2^{K(\rho-1)} + o(1) > 0$, and*

$$\limsup_{\mathcal{N}\to\infty} \frac{1}{\mathcal{N}} \left| \sum_{p<\mathcal{N}} \mu(\mathcal{N}-p)\log p \right| \le \sqrt{C_{K,\rho}/b} + \limsup_{\mathcal{N}\to\infty} \max_{1\le j\le b} \frac{|\sum_{p<\mathcal{N}} \mu(\mathcal{N}-p) W_j(\mathcal{N}-p)\log p|}{m_j \mathcal{N}}. \tag{8.1}$$

The constant $C_{K,\rho}$ is independent of b.

Proof. Let P_0 be the product of the list primes for a fixed copy. For every $d \mid P_0$,

$$\frac{1}{\varphi(P_0)} \sum_{\substack{a \bmod P_0 \\ (a,P_0)=1}} \mathbf{1}_{d\mid \mathcal{N}-a} = \frac{1}{\varphi(d)}. \tag{8.2}$$

This follows because $\mathcal{N}$ is a unit modulo d, so the divisibility condition selects one reduced class. It does not require translation to permute all reduced classes. Thus the mean (2.14) and second moment (2.15) of LWW, p. 7, apply. The truncated divisor expansion and Bombieri–Vinogradov for $p \equiv \mathcal{N} \pmod d$, $(d,\mathcal{N}) = 1$, give the same weighted moments over primes. All moduli are odd squarefree products of list primes, and the maximum over reduced classes makes the estimate uniform in $\mathcal{N}$.

For the weight truncation write $R_w = \lceil \log L \rceil$, as in LWW Section 2.3, p. 9. Its integer-average error in this range is

$$O(L^{B+1}/R_w!) = o(1), \qquad R_w! = L^{(1+o(1))\log\log L}. \tag{8.3}$$

This distinguishes R_w from the moment order R. The variance argument (2.16)–(2.19), pp. 7–10, is applied to $b^{-1}\sum_j W_j/m_j$. Cauchy–Schwarz against the bounded target and the triangle inequality give (8.1). The means tend to the same positive fixed constant, so normalization has a bounded cost independent of b. For each fixed b, let $\mathcal{N}$ tend to infinity before letting b tend to infinity; thresholds may depend on its chosen bands. □

9 Squarefree values

Proposition 9.1 (Uniform squarefree density)**.** *For even $\mathcal{N} \to \infty$,*

$$\sum_{\substack{p<\mathcal{N} \\ p\nmid\mathcal{N}}} \mu^2(\mathcal{N}-p)\log p = A_{\mathcal{N}}\mathcal{N} + O_A(\mathcal{N}L^{-A}), \qquad A_{\mathcal{N}} \ge 0.3739. \tag{9.1}$$

Consequently

$$\#\{p < \mathcal{N} : \mu^2(\mathcal{N}-p) = 1\} = (A_{\mathcal{N}} + o(1))\pi(\mathcal{N}), \tag{9.2}$$

uniformly in even $\mathcal{N}$.

Proof. Use $\mu^2(n) = \sum_{d^2 \mid n} \mu(d)$. If $p \nmid \mathcal{N}$ and $d^2 \mid \mathcal{N} - p$, then $(d, \mathcal{N}) = 1$. For $d \le L^{A+2}$, Siegel–Walfisz in the reduced class $p \equiv \mathcal{N} \pmod{d^2}$, at a sufficiently large fixed precision, gives main term

$$\mathcal{N} \sum_{(d,\mathcal{N})=1} \frac{\mu(d)}{\varphi(d^2)} = A_{\mathcal{N}} \mathcal{N}, \tag{9.3}$$

with extension error $O(\mathcal{N} L^{-A-1})$. The tail has the finite range required by $\mathcal{N} - p > 0$:

$$\sum_{L^{A+2} < d \le \sqrt{\mathcal{N}}} (\mathcal{N}/d^2 + 1) L \ll \mathcal{N} L^{-A-1} + \sqrt{\mathcal{N}} L. \tag{9.4}$$

The last term is absorbed in $O_A(\mathcal{N} L^{-A})$. This is LWW Proposition 6.1 and (6.3), pp. 46–47, with the size and reduced-class checks for the varying shift supplied here. Primes dividing $\mathcal{N}$ cost $O(\omega_{\mathrm{all}}(\mathcal{N}) L) = O(L^2)$. To pass to ordinary counts, use Lemma 9.2 below with the bounded target $\mu^2(\mathcal{N} - p)$. The same progression proof permits any endpoint $y \le \mathcal{N}$. □

Lemma 9.2 (Removing the prime logarithm)**.** *For any bounded target $|f(p)| \le 1$ on primes below $\mathcal{N}$, possibly depending on $\mathcal{N}$, and fixed $B > 2$,*

$$\sum_{p < \mathcal{N}} f(p) = \frac{1}{L} \sum_{p < \mathcal{N}} f(p) \log p + O_B\left(\frac{\mathcal{N} \log L}{L^2} + \frac{\mathcal{N}}{L^B} \right). \tag{9.5}$$

Proof. For $p > \mathcal{N}/L^B$, $\log p / L = 1 + O_B(\log L / L)$. Chebyshev's bound gives $O(\mathcal{N}/L)$ such primes. For the remaining primes bound both sums by their number, at most $\mathcal{N}/L^B$; this proves the identity without varying the target sequence in a partial-summation limit. □

10 Assembly of the Goldbach statements

Proofs of Theorems 1.1 and 1.2. Choose ρ, $A^* > 0$ and then $B > A^* + 3$. Let C_{coef} be the corresponding fixed coefficient and smoothing-loss exponent for this choice of B. Complete the parameter choices in the following order:

$$\begin{gathered} \rho \ ; \ A^* \ ; \ B \ ; \ C_{\mathrm{coef}} \ ; \\ D^* > 2(A^* + B + C_{\mathrm{coef}} + c_*) \ ; \ E_0 \ ; \ G \ ; \\ A_0 \ ; \ K \ ; \ C^*, C^\dagger, C_0 \ ; \ B_0, B_1 \ ; \\ A_{\mathrm{energy}} = D^* + O(A_0 + B_1) \ ; \\ D, C_{\mathrm{zf},1}, C_{\mathrm{zf},2}, C_3, A_s, A_1 \ ; \\ \text{the fixed precision of Proposition 2.2} \ ; \ a_i \ ; \ L_0. \end{gathered} \tag{10.1}$$

Here c_* is a fixed reserve for scale and dyadic losses, chosen large enough before the later parameters. The choices for the moment estimate are those in LWW Section 4.7, p. 39; A_0, K are independent of the band locations. The zero-free constants are subsequently defined by (6.44) with $A = A_{\mathrm{energy}}$ and do not enter C_{coef} or the inequality for D^*. For any fixed b the bK band exponents can be selected after these saving exponents; their positions affect thresholds, not these choices.

On each of the $O(\log L)$ dyadic prime intervals, the Vaughan reduction and Proposition 7.1 reduce the weighted prime sum to Type II sums T_Y on $O(L^{0.2})$ product scales. Propositions 5.3 and 6.1, followed by Theorem 6.12, give

$$Q_Y^{\mathrm{all}} = Q_Y^{\mathrm{maj},\mathcal{N}} + O(\mathcal{N} Y L^{-D^*}), \quad |Q_Y^{\mathrm{maj},\mathcal{N}}| \ll \mathcal{N} Y L^{-D^*}, \quad |T_Y|^2 \ll (\mathcal{N}/Y) Q_Y^{\mathrm{all}} \ll \mathcal{N}^2 L^{-D^*}. \tag{10.2}$$

Thus $|T_Y| \ll \mathcal{N}L^{-D^*/2}$, and the larger fixed budget in (10.1), using $\mathcal{N}/X \ll 2L^B$, gives $T_3(F) \ll xL^{-A^*}$. For Theorem 1.1, use the selected H1*; (1.2) supplies it for every fixed selection of constants. For Theorem 1.2, use Corollary 6.15 with $A_1 = A_0 + C + 1$ and the Burgess input in Theorem 6.12. In either case

$$\sum_{x<p\le 2x} (\log p)\mu(\mathcal{N}-p)W(\mathcal{N}-p) = O(xL^{-A^*}), \qquad \sum_{p<\mathcal{N}} (\log p)\mu(\mathcal{N}-p)W(\mathcal{N}-p) = o(\mathcal{N}). \tag{10.3}$$

The omitted small primes have the negligible bounded-weight cost stated in Remark 2.2.

For each fixed b, every normalized weighted limit in (8.1) is zero. Then let $b \to \infty$ to get $\sum_{p<\mathcal{N}} \mu(\mathcal{N}-p)\log p = o(\mathcal{N})$. Lemma 9.2 and $\pi(\mathcal{N}) \sim \mathcal{N}/L$ give the initial conclusion of (1.4). Finally, for $\kappa = \pm 1$,

$$\#\{p < \mathcal{N} : \mu(\mathcal{N}-p) = \kappa\} = \tfrac{1}{2}\left(\#\{p < \mathcal{N} : \mu^2(\mathcal{N}-p) = 1\} + \kappa \sum_{p<\mathcal{N}} \mu(\mathcal{N}-p)\right). \tag{10.4}$$

Proposition 9.1 proves the sign densities.

All comparisons use $\log x = L + O_B(\log L)$ on the stated range. Puncturing, the large common-prime discard, the power errors in Lemma 4.1 and the $O(\log L)$ dyadic summation are uniform in even $\mathcal{N}$. Constants depend on the fixed bands, K, ρ, b, and requested precisions, and in the small-radical case on C. The limits are ineffective through Siegel–Walfisz and Proposition 2.2. A fixed H1* supplies the selected fixed energy precision; the half-plane and the small-radical region supply every fixed precision needed in this order. □

11 Small shifts

Throughout this section x is the large parameter, $L = \log x$, and $0 < |h| \le L^C$, with fixed C. All constants are uniform in this range. We use the prime form $p + h$ on the domain $p + h \ge 1$.

11.1 Setting and reductions

Define $F_h(n) = \mu(n+h)W(n+h)$ when $n + h \ge 1$, and 0 otherwise. On $(x, 2x]$ the arguments are positive for large x. Puncture the bands at $p \mid h$, costing at most $\omega_{\mathrm{all}}(h)e^{-L^{0.1}}$ in reciprocal mass. Every prime divisor of h is below Z, so the sifted Vaughan variables satisfy $(mnrs, h) = 1$ automatically. Write $mn + h = ah'$, $a \in \mathcal{A}$. A common prime of a, m would divide h, which puncturing excludes. Remove $(a, h') > 1$ as in LWW (4.3), p. 15. Now $mn + h \asymp X$ on every rectangle, so the common cofactor range is $O(X/Y)$ and

$$|T_Y|^2 \ll (X/Y)Q_Y, \qquad Q_Y \ge 0. \tag{11.1}$$

The Lemma 4.1 check is $d \mid \ell h' + h \Rightarrow (d, \ell) \mid h$ and $(\ell, h) = 1$, so $(d, \ell) = 1$. For $(a, b) = 1$ the equation is

$$b(mn+h) = a(rs+h) \quad \Longleftrightarrow \quad t = bmn - ars = -h(b-a), \qquad h \mid t, \quad t'' = -t/h = b - a. \tag{11.2}$$

It is LWW (4.8), p. 16, when $h = 2$.

11.2 Oriented bases, parity and root counts

Lemma 11.1 (Small-shift blocks and orientation)**.** *Use labelled lattices Λ_ϱ of index $|h|$ for unit slopes modulo $|h|$ and odd-odd endpoint positions. For a primitive odd-odd root $P = (u, v)^T$ with $(uv, h) = 1$, choose $ud - vc = 1$, $0 \le c < u$, $r_P = c/u$. Put*

$$\begin{aligned} &P^* = -h(c, d)^T, \qquad g_h(P) = \begin{pmatrix} u & -hc \\ v & -hd \end{pmatrix} = \begin{pmatrix} u & -hur_P \\ v & -h(1/u + vr_P) \end{pmatrix}, \\ &E_h = \operatorname{diag}(1, -h), \quad \det g_h(P) = -h, \quad \tau_h(z) = z_1 - hr_P z_2. \end{aligned} \tag{11.3}$$

This is a basis of the index-$|h|$ lattice. For $Q = g_h(P)z$,

$$\det(P, Q) = -hz_2, \qquad Q_1 = u\tau_h(z), \qquad Q_2 = v\tau_h(z) - hz_2/u. \tag{11.4}$$

Lattice-primitivity is $(z_1, z_2) = 1$; integer-plane primitivity also requires $(Q_1, h) = 1$. For primitive z and $p \nmid h$, list divisibility is $[z]_p = \ker(u, -hc) \bmod p$. For $\det(z, w) = 1$, the intrinsic ratio obtained with a complement of determinant $|h|$ is

$$\beta_Q^{(h)} \equiv -\operatorname{sgn}(h)\frac{\tau_h(w)}{\tau_h(z)} \pmod 1, \qquad \partial_r \frac{\tau_h(w)}{\tau_h(z)} = -\frac{h}{\tau_h(z)^2}. \tag{11.5}$$

Its absolute derivative lies in $[|h|/256, 256|h|]$. The odd-odd restriction is

$$\begin{cases} z_1 \text{ odd}, & h \text{ even}, \\ z_1 \text{ odd and } z_2 \text{ even}, & h \text{ odd}. \end{cases} \tag{11.6}$$

Proof. The columns lie in Λ_ϱ and have determinant $-h$, so the basis and coordinate assertions follow with that orientation. The first-row kernel assertion follows from invertibility modulo $p \nmid h$. If $W = g_h(P)w$ has relative determinant 1, then $\det(Q, W) = -h$. Multiplication of W by $-\operatorname{sgn}(h)$ gives determinant $|h|$, proving (11.5). At an integer-plane primitive state this ratio is $|h|r_Q \pmod 1$. Both good-state tests use distance to integers, so changing the ratio's sign preserves them; absolute-value boundary bounds also preserve their estimates. The derivative formula follows by differentiation, and $\tau_h(z) \in [1/16, 16]$ gives its range, bounded below by $1/256$ and above by $x^{o(1)}$. For even h, $P^* \equiv 0 \pmod 2$, giving the initial parity case. For odd h, $P^* \equiv (c, d) \pmod 2$ and $d - c \equiv ud - vc \equiv 1 \pmod 2$. Hence $Q \equiv (z_1 + cz_2, z_1 + dz_2) \pmod 2$, proving the other case. □

Let $\mathfrak{M}_h = \{\theta : |h|\theta \in \mathfrak{M}_1\}$. The covering identity is

$$\int_{\mathfrak{M}_h} e(\theta(t + h(b - a)))\, d\theta = \mathbf{1}_{h|t} \int_{\mathfrak{M}_1} e(\phi(t'' - b + a))\, d\phi. \tag{11.7}$$

Indeed the branch substitution first gives phase $e(-\operatorname{sgn}(h)\phi(t'' - b + a))$; for $h > 0$ reflect ϕ, using symmetry of $\mathfrak{M}_1$. The kernel is block diagonal and its reduced determinant is exactly $\det(z, z')/D$. A full block can contain $\asymp d_0 YMN/|h|$ positions, so Lemma 3.4's $O(d_0 Y)$ application bound is replaced by the local partition below.

Parity is imposed only through the multiplier χ in LWW (4.34), p. 25, and the corresponding sublattice restriction in M_R: use (11.6) on the full relative integer lattice. This projection has norm at most 1. The labelled enlargement is used only for nonunit states modulo h, where the endpoint vector vanishes. In particular even positions are not separately reintroduced this way. For odd h, allowed reduced determinants are even, which is just a restriction of this projection. The endpoint multiplier $\chi(B_D k)$ in (4.45), p. 29, and compression of the clean-edge operator cannot increase its norm.

Lemma 11.2 (Lemma R for small shifts)**.** *Let $d_0YM, N \ge x^\delta$ and let S be odd squarefree, $(S,h)=1$, $S \le \exp(C_0 L^{0.98})$. Use the intervals I_1, I_2, I_3 of Lemma 4.1. For a matrix $g_0 \bmod S$ of determinant $-h$, the number of primitive roots with u, v odd, $(uv,h)=1$, $u/(d_0YM) \in I_1$, $v/N \in I_2$, $r_P \in I_3$ and $g_h(P) \equiv g_0 \pmod S$ is*

$$\frac{c_h d_0 YMN|I_1||I_2||I_3|}{|\mathrm{SL}_2(\mathbb{Z}/S\mathbb{Z})|} + O(d_0YMNx^{-c\delta}), \qquad c_h = \frac{1}{\zeta(2)} \prod_{p \mid 2h} \frac{p-1}{p+1}. \tag{11.8}$$

Prescribed lines have weights $1/(p+1)$ and a uniform error of the same form, with a possibly smaller fixed positive c. For $h = \pm 2$, $c_h = 1/(3\zeta(2))$.

Proof. Use LWW Lemma A.1 and (A.5), pp. 47–51, with the original ratio $r_P = c/u$. Since $(h,S)=1$, $g_h \equiv g_0 \pmod S$ is equivalent to $g \equiv g_0 E_h^{-1} \pmod S$. The odd-odd condition supplies the original parity factor at 2. Sieve the odd part of $\mathrm{rad}(h)$ on both coordinates as in Lemma 4.1; its local factor is $(p-1)/(p+1)$ and its divisor cost $2^{\omega_{\mathrm{all}}(h)} \le L^{O(1)}$. The swapped-coordinate case is LWW's determinant-one calculation with displacement $1/(uv)$, since the counted ratio remains r_P. Thus there is no $\mathcal{N}/(uv)$ displacement to discard. The ratio map $r \mapsto |h|r$ preserves Haar measure, and the line distribution follows from the determinant-$(-h)$ matrix coset just as in (4.3). □

11.3 Locality within a small-shift block

Proposition 11.3 (Small-shift Type II estimate)**.** *Using* H2′ *and the changes below,*

$$|Q_Y - Q_Y^{\mathrm{maj},h}| \ll_A XYL^{-A} \tag{11.9}$$

uniformly for $0 < |h| \le L^C$.

Proof. The relative kernel has $t'' = \det(z,z')/D$ and the same list structure. Rows 5, 9 and 17 of Table 2 are replaced as follows; all support-dependent constants refer to the fixed cutoff ψ_1. On an edge,

$$\begin{aligned} |\det(P_j, P_{j+1})| &\ll |h| d_0 Y, \\ |\operatorname{slope}(P_{j+1}) - \operatorname{slope}(P_j)| &\ll \frac{|h|}{d_0 Y M^2}. \end{aligned} \tag{11.10}$$

Summing at most $2R$ edges from $P = P_0$ gives

$$|\det(P, P_j)| \ll R|h| d_0 Y, \quad |(z_j)_2| \ll R d_0 Y, \quad |(z_j)_1| = |\tau_h(z_j) + h r_P (z_j)_2| \ll |h| R d_0 Y = x^{o(1)}. \tag{11.11}$$

Use the relative cube Z_0 of radius $2C_{\mathrm{pos}}|h|Rd_0Y$ in (4.33), p. 24. Its coefficient mass remains $\exp(O_K(L^{0.73}))$, since $\log(|h|Rd_0Y) = O(L^{0.21})$. The second position boundary, with its orientation retained, is

$$E_2 = t_0 \tau_h(z) - \frac{h z_2}{uN}, \qquad \left| \frac{h z_2}{uN} \right| \ll \frac{R|h|}{MN}. \tag{11.12}$$

Its derivatives have the bounds of LWW (4.30), pp. 23–24.

Inside each labelled block partition the physical slope P_2/P_1 into half-open intervals of length $|h|/(d_0YM^2)$. By (11.10), each interval interacts with $O(1)$ neighbors in that block. Each contains $O(|h|d_0Y)$ lattice-primitive positions: choose one such P in it. Other positions have $|\det(P,Q)| \ll |h|d_0Y$ and determinants in $|h|\mathbb{Z}$. For a fixed determinant their differences belong to $\mathbb{Q}P \cap \Lambda_\varrho = \mathbb{Z}P$,

by lattice-primitivity, so the first-coordinate strip gives $O(1)$ choices. This reasoning also applies to added nonunit states and does not assume their primitivity in $\mathbb{Z}^2$. The trace factor is

$$(|h|d_0Y)^{1/(2R)} = \exp(O(L^{0.21}/L^{1/2})) = O(1). \tag{11.13}$$

For the comparison count, $h \mid bk_0 - a\ell_0$ and $|bk_0 - a\ell_0| \ll |h|Y$, giving $O(1)$ choices of ℓ_0 in one class modulo $|h|$ in an interval of length $O(|h|)$. For endpoint primitivity, $p \mid (m, s)$ would give $p \mid h(b-a)$ with $p > Z > |h|$ and $0 < |b-a| < 4Y < Z$, a contradiction. The other dictionary rows, Lemma 11.2, and LWW's arguments then give (11.9). □

11.4 Small-shift major arcs

Proposition 11.4 (Small-shift major term)**.** *Using the arguments of LWW Section 5 as written,*

$$|Q_Y^{\mathrm{maj},h}| \ll_D XYL^{-D} \tag{11.14}$$

uniformly for $0 < |h| \le L^C$.

Proof. The kernel from (11.7) carries

$$\mathbf{1}_{bmn\equiv ars\ (h)}\psi_1(t''/Y)\int_{\mathfrak{M}_1} e(\phi(t''-b+a))\,d\phi.$$

Now $\psi_1(t''/Y) = \psi_1((ars-bmn)/(hY))$ forces $|bmn-ars| \ll |h|Y \ll bmn$, a genuine localization. Apply the proof of LWW Proposition 5.3, pp. 42–45, with these changes. The congruence expands as

$$\mathbf{1}_{bmn\equiv ars\ (h)} = \frac{1}{\varphi(|h|)}\sum_{\chi \bmod |h|}\chi(bmn)\overline{\chi(ars)}. \tag{11.15}$$

On $\phi = h_1/k + \lambda_0/Y$,

$$e(h_1t''/k) = e(-h_1bmn/(hk))e(h_1ars/(hk)).$$

Expand these unit-residue functions factor by factor in characters modulo $|h|k$, and the factors $e(\mp h_1(b-a)/k)$ modulo k. The total character mass is at most $(|h|k)^6 \le L^{6(A_0+C)}$.

Mellin separation uses $|h|Y$ in place of Y and reflects the cutoff when its orientation requires it. Its relative scale is $|h|/X$, giving

$$|t| \le (X/|h|)L^{B_1} \le XL^{B_1}, \qquad T_* = \exp(L/(\log L)^2). \tag{11.16}$$

These are in the range of LWW Lemmas 5.1–5.2, pp. 40–41. The kernel has X replaced by X/h; its prefactor gains at most an additional $|h| \le L^C$. Fix one character term. Its conductor is at most $|h|k \le L^{A_0+C}$. LWW (5.14), p. 43, follows with arbitrary fixed precision A from (5.2), p. 40, Lemma 5.2 and Proposition 2.2 at the fixed modulus exponent $A_0 + C$. Lemma 5.2 is uniform in the bounded character twist and gives an energy estimate. The arbitrary saving comes from Proposition 2.2. To absorb the character mass and the prefactor choose

$$A = D + 9A_0 + 7C + B_1 + 1 \tag{11.17}$$

in place of LWW's $D + 9A_0 + B_1 + 1$. The rest of their assembly is used as written. All moduli are $L^{O(1)}$, so this step uses no H1 or H1*. It uses LWW Section 5 in addition to H2$'$. □

11.5 Small-shift Type I

Proposition 11.5 (Small-shift Type I)**.** *LWW Section 3 with the following divisor and parity substitutions gives $T_1(F_h), T_2(F_h) \ll_A xL^{-A}$ uniformly in h.*

Proof. The sifted q satisfies $(q,h)=1$. After expanding R_Z, let $g=(r,h)$ with r odd squarefree. The list divisor d is disjoint from rh. Apply (7.2) and write

$$qrj+h=dgat, \qquad t\equiv (h/g)(da)^{-1} \pmod{qr/g}, \qquad t\geq 1. \tag{11.18}$$

The inverse exists because $(da, qr/g)=1$. The class is reduced since $(h/g, r/g)=(h,q)=1$ and $(g, qr/g)=1$. The primes of a may lie in g, so no term is removed by a condition $(g,da)=1$. The Möbius function is evaluated at t itself in this substitution; factoring out g after evaluating it would instead require $\mu(gt')=\mu(g)\mu(t')\mathbf{1}_{(t',g)=1}$.

For even h, j odd is equivalent to t odd, and the class modulo $2qr/g$ is reduced. For odd h, j odd is equivalent to t even. Put $t=2t''$; then

$$t''\equiv (h/g)(2da)^{-1} \pmod{qr/g}, \qquad \mu(2t'')=-\mu(t'')\mathbf{1}_{2\nmid t''}. \tag{11.19}$$

The last indicator makes the class modulo 2 reduced, so in both cases the Möbius sum is over a reduced class modulo $2qr/g$. Apply (3.1) with the weighted grouping of LWW p. 13 separately at each g-dependent level $x/(dg)$ (and $x/(dga)$ in the small-a part). Then sum $\sum_{g\mid \mathrm{rad}(h)} 1/g \ll \log L$. The extra Rankin factor $\prod_{p\mid g}(1-p^{-1/20})^{-1}\leq L^{O(1)}$ is absorbed by $x^{-1/50}$ as in (7.5). The positive interval for $p+h\in(x+h, 2x+h]$ is handled by the maximum over endpoints in the same estimate. □

Remark 11.6 (Other Type I parametrizations). For comparison, factoring $t=gt'$ after the original substitution requires the factor $\mathbf{1}_{(t',g)=1}$ in $\mu(gt')=\mu(g)\mu(t')\mathbf{1}_{(t',g)=1}$. One may split into at most $\varphi(g)\leq |h|$ reduced classes modulo qr and absorb L^C by increasing the distribution precision. Another exact expansion is

$$R_Z(k)=\mathbf{1}_{(k,h)=1}R'_Z(k), \qquad R'_Z(k)=\mathbf{1}_{\text{each prime divisor of } k \text{ exceeds } Z \text{ or divides } h}.$$

Expanding only R'_Z leaves r coprime to h; the remaining condition on j can be expanded over $e\mid \mathrm{rad}(h)$, with $2^{\omega_{\mathrm{all}}(h)}\leq L^{O(1)}$ terms. The proof above uses (11.18) instead.

11.6 Removal of weights and assembly

Proof of Theorem 1.3. For weight removal, $d\mid P_0$ and $(h,d)=1$ give

$$\frac{1}{\varphi(P_0)}\sum_{\substack{a \bmod P_0\\(a,P_0)=1}}\mathbf{1}_{d\mid a+h}=\frac{1}{\varphi(d)}. \tag{11.20}$$

Bombieri–Vinogradov uses the reduced class $p\equiv -h \pmod d$. Use the same mean normalization as in Proposition 8.1.

For squarefree density use $1\leq p+h\leq x+|h|$. If $p\nmid h$ and $d^2\mid p+h$, then $(d,h)=1$; otherwise a common prime divisor would equal $p\mid h$, which accounts for at most $\omega_{\mathrm{all}}(h)$ primes. Siegel–Walfisz applies uniformly to the reduced class $p\equiv -h \pmod{d^2}$ for $d\leq L^{A+2}$. For the tail, restricting to $d\leq\sqrt{x+|h|}$ gives

$$L\sum_{d>L^{A+2}}\frac{x+|h|}{d^2}+O(\sqrt{x+|h|}\,L), \tag{11.21}$$

and the extra endpoint term is absorbed at any fixed precision. Thus

$$\#\{p \le x : p+h \ge 1,\ \mu^2(p+h)=1\} = (A_h + o(1))\pi(x), \qquad A_h \ge 0.3739, \tag{11.22}$$

uniformly in h.

Assemble Propositions 11.3, 11.4 and 11.5, using $|T_Y|^2 \ll (X/Y)Q_Y$ and $Q_Y \ll XYL^{-D^*}$. The dyadic summation and normalized weight removal give the weighted cancellation, and the bounded-target comparison (9.5), with $\mathcal{N}$ replaced by x, gives its unweighted form. The negligible lower prime range includes all $p+h \le 0$, at most $O(L^C)$ primes; the domain in (1.5) excludes them throughout. When summing dyadic intervals with lower endpoint $\gg x/L^B$, their logarithms are $L + O_B(\log L)$, so the fixed-C range and all bounds above remain uniform. Apply the sign identity (10.4) on positive arguments to obtain (1.5). This uses LWW Sections 2.3, 3, 5 and 6, as well as H2′, exactly as in the statement. □

12 Liouville versions

The function $\lambda(n) = (-1)^{\Omega(n)}$ is completely multiplicative and takes values in $\{-1,1\}$ for every $n \ge 1$. We retain the Vaughan coefficient $\mu(m)R_Z(m)$, since it comes from decomposing the prime variable. For the shifted target define explicitly

$$\begin{aligned} F_{\lambda,G}(n) &= \begin{cases} \lambda(\mathcal{N}-n)W(\mathcal{N}-n), & 1 \le n < \mathcal{N}, \\ 0, & \text{otherwise}, \end{cases} \\ F_{\lambda,h}(n) &= \begin{cases} \lambda(n+h)W(n+h), & n \ge 1,\ n+h \ge 1, \\ 0, & \text{otherwise}. \end{cases} \end{aligned} \tag{12.1}$$

For $\kappa = \pm 1$, the three statements of Theorem 1.4 are explicitly

$$\sum_{p<\mathcal{N}} \lambda(\mathcal{N}-p) = o(\pi(\mathcal{N})), \qquad \#\{p<\mathcal{N} : \lambda(\mathcal{N}-p)=\kappa\} = \left(\tfrac{1}{2}+o(1)\right)\pi(\mathcal{N}), \tag{12.2}$$

$$\sum_{\substack{p\le x \\ p+h\ge 1}} \lambda(p+h) = o(\pi(x)), \qquad \#\{p \le x : p+h \ge 1,\ \lambda(p+h)=\kappa\} = \left(\tfrac{1}{2}+o(1)\right)\pi(x). \tag{12.3}$$

The former holds for all even $\mathcal{N}$ under the hypotheses of Theorem 1.1, and for the fixed-C small-radical class under those of Theorem 1.2; the latter is uniform for $0 < |h| \le (\log x)^C$ under those of Theorem 1.3. LWW Section 1, pp. 1–3, mentions the Liouville counterpart; we supply the expansion and Type I substitutions needed for it.

12.1 Expansion of the weight

The exact identities are

$$\begin{aligned} \omega_i(n)\rho^{\omega_i(n)-1} &= \sum_{\substack{p\in\mathcal{P}_i \\ p\mid n}} \rho^{\omega_i(n)-1}, \\ W(n) &= \frac{1}{\prod_i V_i} \sum_{\substack{a\in\mathcal{A} \\ a\mid n}} \rho^{\omega(n)-K}, \\ \omega(ah)-K &= \omega(h)-s(a,h), \qquad s(a,h) = \#\{i : p_i \mid h\} \in \{0,\dots,K\}, \end{aligned} \tag{12.4}$$

where $\omega_i(ah) = \omega_i(h) + \mathbf{1}_{p_i\nmid h}$ and $s(a,h) = 0$ precisely when $(a,h)=1$.

Lemma 12.1 (Liouville expansion). *Let w denote $mn+2$, $\mathcal{N}-mn$ or $mn+h$, with the positive argument convention used above and the corresponding coefficients. Then*

$$\sum_{\substack{m,n\\ w\ge 1}} \alpha_m\beta_n\lambda(w)W(w) = \frac{(-1)^K}{\prod_i V_i}\sum_{\ell\ge 1}\lambda(\ell)\rho^{\omega(\ell)} \sum_{\substack{a\in\mathcal{A},m,n\\ w=a\ell,(a,\ell)=1}} \alpha_m\beta_n + O_{K,\rho}(XLe^{-L^{0.1}}). \tag{12.5}$$

For a fixed positive w, the discrepancy before multiplication by $\alpha_m\beta_n$ is exactly

$$\frac{\lambda(w)}{\prod_i V_i}\sum_{\substack{a\in\mathcal{A},a\mid w\\ s(a,w/a)\ge 1}} \rho^{\omega(w)-K}. \tag{12.6}$$

It lies in $\lambda(w)[0,W(w)]$; every summand weight is at most 1.

Proof. For $a\in\mathcal{A}$, $\lambda(a)=(-1)^K$, and complete multiplicativity gives $\lambda(w)=(-1)^K\lambda(w/a)$ for every $a\mid w$, with no coprimality assumption. On $(a,w/a)=1$, (12.4) gives exactly the main term. The omitted summands are precisely (12.6). They have some $p_i^2\mid w$, and $\omega(w)\ge K$ makes their weights at most 1. Their absolute value is bounded, with constant 1, by

$$\frac{1}{\prod_i V_i}\sum_{\substack{a\in\mathcal{A},a\mid w\\ (a,w/a)>1}} 1.$$

Since $W(w)\ll_{K,\rho} 1$, the absolute discrepancy is at most

$$C_{K,\rho}\sum_{p\in\cup_i\mathcal{P}_i}\mathbf{1}_{p^2\mid w}.$$

For fixed m on the coefficient support, $(m,p)=1$ and the condition $p^2\mid w$ selects one class of n modulo p^2. Since $N\gg p^2$, its count is at most $N/p^2+1\ll N/p^2$. Summing with $|\alpha_m|\le 1$, $|\beta_n|\ll L$ and $\sum_{p\in\cup_i\mathcal{P}_i}p^{-2}\ll e^{-L^{0.1}}$ gives the claimed error for all three positive shifted arguments. This is the bounded-weight square-divisibility argument of LWW (4.3), p. 15.

If the restriction $(a,\ell)=1$ is then removed before Cauchy, the added terms have $s\ge 1$ and $\rho^{\omega(\ell)}\le\rho^{\omega(w)-K}$, by (12.4). They are bounded by the same count, at an additional fixed constant. Cauchy uses only $|\lambda(\ell)\rho^{\omega(\ell)}|\le 1$. For the Goldbach shift use $Q_Y^+\le Q_Y^{\mathrm{all}}$ in (2.14); for small shifts use (11.1). In each case the resulting square contains no target function. □

12.2 Bombieri–Vinogradov for Liouville

The Möbius input is LWW (3.1), p. 10: for every $A>0$ there exists $B=B(A)$ such that

$$\sum_{q\le x^{1/2}L^{-B}}\max_{(c,q)=1}\max_{y\le x}\left|\sum_{\substack{n\le y\\ n\equiv c\ (q)}}\mu(n)\right|\ll_A xL^{-A}. \tag{12.7}$$

Lemma 12.2 (Liouville distribution). *For every $A>0$ there is $B'(A)$ such that*

$$\sum_{q\le x^{1/2}L^{-B'(A)}}\max_{(c,q)=1}\max_{y\le x}\left|\sum_{\substack{n\le y\\ n\equiv c\ (q)}}\lambda(n)\right|\ll_A xL^{-A}. \tag{12.8}$$

One may take $B'(A)=\max(B(A),A)+A+1$.

Proof. The multiplicative identity

$$\lambda(n) = \sum_{d^2 \mid n} \mu(n/d^2) \tag{12.9}$$

holds prime by prime: at p^k the sum is 1 for even k and -1 for odd k. For $(c, q) = 1$,

$$\sum_{\substack{n \le y \\ n \equiv c \ (q)}} \lambda(n) = \sum_{d \le \sqrt{y}} \sum_{\substack{m \le y/d^2 \\ d^2 m \equiv c \ (q)}} \mu(m).$$

There is no solution when $(d, q) > 1$; otherwise the class is $m \equiv c\bar{d}^2 \pmod q$, which is reduced. Put $D_0 = L^{A+1}$ and $Q = x^{1/2} L^{-B'}$. For $d \le D_0$, use the discrepancy

$$E_\mu(x/d^2; q) = \max_{(c',q)=1} \max_{z \le x/d^2} \left| \sum_{\substack{m \le z \\ m \equiv c' \ (q)}} \mu(m) \right|.$$

For $d > D_0$ use $y/(d^2 q) + 1$. The tail summed over $q \le Q$ is

$$\ll xL/D_0 + Q\sqrt{x} \le xL^{-A} + xL^{-B'}. \tag{12.10}$$

For the head, $x_d = x/d^2 \ge xL^{-2A-2}$ and $\log x_d \ge L/2$. Condition $B' \ge B(A) + A + 1$ ensures $Q \le \sqrt{x_d} (\log x_d)^{-B(A)}$. Apply (12.7) at x_d and sum $\sum_{d \le D_0} x d^{-2} L^{-A} \ll_A xL^{-A}$. The stated $B'(A)$ also absorbs the tail. □

Lemma 12.3 (Weighted Liouville distribution)**.** *Put*

$$E_\lambda(Y; m) = \max_{(c,m)=1} \max_{z \le Y} \left| \sum_{\substack{t \le z \\ t \equiv c \ (m)}} \lambda(t) \right|.$$

For fixed $A_{\mathrm{BV}} > 2A + 4$ and $Q \le Y^{1/2} L^{-B'(A_{\mathrm{BV}})}$,

$$\sum_{m \le Q} \tau(m) E_\lambda(Y; m) \ll_A YL^{-A}. \tag{12.11}$$

Here $\log Y \asymp L$ in the applications, with fixed comparison constants and a corresponding fixed precision margin.

Proof. The trivial bound is $E_\lambda(Y; m) \ll Y/m + 1$. As in the Cauchy step of LWW Section 3, p. 13,

$$\Big(\sum \tau E\Big)^2 \le \Big(\sum E\Big)\Big(\sum \tau^2 E\Big), \quad \sum E \ll YL^{-A_{\mathrm{BV}}}, \quad \sum \tau^2 E \ll YL^4 + QL^3. \tag{12.12}$$

Use Lemma 12.2 at precision A_{BV}, rather than A, and take square roots. □

12.3 Type I for Liouville

Proposition 12.4 (Liouville Type I). *With the corresponding positive-argument functions F_λ, $T_1(F_\lambda), T_2(F_\lambda) \ll_{A,K,\rho} xL^{-A}$ for shift $+2$, Goldbach shift $\mathcal{N}$, and small shift h under the same uniformity ranges as above.*

Proof. The truncated inclusion–exclusion for R_Z, divisor expansion of W and truncation error (3.3) in LWW Section 3, pp. 10–12, use only $|\alpha_q| \le L$, boundedness of F, and counts in residue classes. For shift $+2$ the remaining expression is

$$\sum_q \alpha_q \sum_r \mu(r) \sum_{\substack{d \\ (d,r)=1}} c_d \sum_{\substack{x<qk\le y \\ 2\nmid k, r\mid k \\ d\mid qk+2}} \lambda(qk+2), \qquad y \in [x, 2x]. \tag{12.13}$$

Write $k = rj$. Now d odd and $d \mid qrj+2$ imply $(d, qr) = 1$. Replace (3.4) by complete multiplicativity:

$$\lambda(dv) = \lambda(d)\lambda(v), \qquad qrj + 2 = dt, \qquad \lambda(d) \sum_{\substack{(x+2)/d<t\le(y+2)/d \\ dt\equiv qr+2 \ (2qr)}} \lambda(t). \tag{12.14}$$

The class is reduced since $(d, 2qr) = (qr+2, 2qr) = 1$. There is no auxiliary a-sum. The divisor satisfies $d \le \exp(O_K(R_I L^{0.2})) = x^{o(1)}$, where $R_I = \lceil C' \log L \rceil$ is the Type I truncation depth. For large x this gives $d \le x^{1/20}$; for the Goldbach substitution below, also $dg \le x^{1/20}$ since $g \le r = x^{o(1)}$. Thus the small-divisor case $da \le x^{1/20}$ with $a = 1$ covers the whole sum, and no Rankin branch is needed.

Group by qr with coefficient at most $L\tau(qr)$. For $Y = (2x+2)/d \ge 2x^{19/20}$ and $2qr \le x^{2/5+o(1)} \le Y^{1/2} L^{-B'(A_{\mathrm{BV}})}$, Lemma 12.3 gives

$$\sum_d |c_d| \sum_{q,r} |\alpha_q| E_\lambda((2x+2)/d; 2qr) \ll xL^{-A} \sum_d |c_d|/d \ll_{K,\rho} xL^{-A}, \tag{12.15}$$

using $\sum_d |c_d|/d \ll_{K,\rho} 1$ from LWW Section 3, p. 11. The T_1 partial summation of p. 14 is unchanged.

For the Goldbach shift, let $g = (r, \mathcal{N})$ and write

$$\mathcal{N} - qrj = dgt, \qquad \lambda(\mathcal{N} - qrj) = \lambda(dg)\lambda(t), \qquad t \equiv (\mathcal{N}/g)d^{-1} \pmod{qr/g}. \tag{12.16}$$

The class is reduced by the coprimality checks in Proposition 7.1, and j odd is equivalent to t odd. Use modulus $2qr/g$ and level $Y = \mathcal{N}/(dg) \ge x^{19/20}$; the modulus is $\le x^{2/5+o(1)}$. Carry out the grouping separately for each g because this level depends on g, then sum $\sum_{g\mid \mathrm{rad}(\mathcal{N})} 1/g \ll \log L$. The endpoints are those of Section 7 with $a = 1$.

For the shift h, put $g = (r, h)$ and use

$$qrj + h = dgt, \qquad \lambda(qrj + h) = \lambda(dg)\lambda(t), \qquad t \equiv (h/g)d^{-1} \pmod{qr/g}. \tag{12.17}$$

For even h the odd t class modulo $2qr/g$ is reduced. For odd h, put $t = 2t''$:

$$\lambda(2t'') = -\lambda(t''), \qquad t'' \equiv (h/g)(2d)^{-1} \pmod{qr/g}. \tag{12.18}$$

There is no parity condition on t'' here. Apply Lemma 12.3 at modulus qr/g and $Y = (2x+h)/(2dg)$; handle each g separately and then sum $1/g$. All cofactors are positive and 0 is excluded. □

12.4 Uses of the target function

Table 1: Target-function substitutions.

Item	LWW input or operation	Liouville substitution
1	(4.3), p. 15; weight expansion	Lemma 12.1, including the square-overlap count.
2	(4.4)–(4.5), p. 15; Cauchy	$\lvert\lambda(\ell)\rho^{\omega(\ell)}\rvert \le 1$; the Goldbach square is (2.14).
3	Sections 4.2–4.7, pp. 17–39; Appendix A, pp. 47–51	The square has no target function; the same moment and root arguments apply.
4	Section 5, pp. 39–45	The Möbius coefficient in M remains $\mu(m)R_Z(m)$ from Vaughan's identity.
5	(3.1), p. 10	Lemma 12.2.
6	(3.4), p. 12, and (3.5), p. 13	$\lambda(dv) = \lambda(d)\lambda(v)$; no a-sum or Rankin branch.
7	Weighted grouping, p. 13	Lemma 12.3, using the higher fixed precision.
8–10	(3.3) truncations; Proposition 2.1; Proposition 2.3 and weight removal	Only boundedness of the target and residue counts; use the same positive means m_j.
11	Proposition 6.1, pp. 46–47	No squarefree-density input is required.
12	Final sign assembly	Use (12.19).

Other Möbius and squarefree factors in LWW occur in the divisor combinatorics of Proposition 2.3 (pp. 7–9), Bombieri–Vinogradov for primes (p. 9), squarefree r, c_d in the Type I expansion (pp. 11–14), the relative-coordinate/list structure (pp. 22–26, 35), the Vaughan coefficient, and the appendix sieve (p. 50). They do not refer to the target sign and are retained. The additional unit discards in Sections 2 and 7 use only boundedness.

12.5 Sign densities and assembly

Proof of Theorem 1.4. For any finite prime set $\mathcal{P}$ and positive $n(p)$,

$$\#\{p \in \mathcal{P} : \lambda(n(p)) = \kappa\} = \tfrac{1}{2}\left(|\mathcal{P}| + \kappa \sum_{p\in\mathcal{P}} \lambda(n(p))\right), \quad \kappa = \pm 1. \tag{12.19}$$

Use $\mathcal{P} = \{p < \mathcal{N}\}$, of size $\pi(\mathcal{N})$ for even $\mathcal{N} > 2$, or $\mathcal{P} = \{p \le x : p + h \ge 1\}$, of size $\pi(x) + O(L^C)$.

In the Goldbach assembly, use Lemma 12.1 and Proposition 12.4 for the expansion and Type I sums. The block estimate, major arcs and Theorem 6.12 are independent of the target. The same normalized weight removal gives the unweighted cancellation. For small shifts use the identical substitutions in Section 11.6. Lemma 12.2 changes the fixed distribution exponent $B(A)$ to $B'(A)$; all choices still precede the large parameter. Uniformity in $\mathcal{N}$ or $|h| \le L^C$ and the respective hypotheses are unchanged. Identity (12.19) proves the sign densities without Proposition 9.1.

On squarefree values $\lambda = \mu$. Thus the nonsquarefree values in the Goldbach setting have density $(1 - A_{\mathcal{N}})/2 + o(1)$ for each Liouville sign, consistent with (1.4) and (12.2). The errors remain ineffective through the cancellation argument. □

A Changes relative to LWW, Sections 4.1–4.7 and Appendix A

All LWW references in Table 2 are to version 1, 14 October 2026. Page numbers are the numbered pages of that paper. Its arguments are used as written with these replacements. The square being

represented is the full all-integer nonnegative square (2.14), whose expansion is

$$Q_Y^{\mathrm{all}} = \prod_i V_i^{-2} \sum_{\substack{a,b,m,n,r,s \\ (\mathcal{N}-mn)/a=(\mathcal{N}-rs)/b\in\mathbb{Z}\setminus\{0\}}} \eta(a/Y)\eta(b/Y)\alpha_m\overline{\alpha_r}\beta_n\overline{\beta_s}. \tag{A.1}$$

After the common-list removal, (2.15) is exactly its determinant condition. Endpoint pairing uses (5.8) with a conjugate-linear initial entry. The moment normalization remains d_0YMN, and Cauchy uses $\mathcal{N}/Y$ separately.

Table 2: The block dictionary.

Row	LWW location	Replacement or retained input
1	Section 4.1, p. 14; (4.3)–(4.5), p. 15	Separate the product cutoff as there, with $F=0$ off positive shifted arguments. Use the nonnegative enlargement (2.14) and its exact integer-cofactor expansion (A.1). Its common range is $\lvert h\rvert \ll \mathcal{N}/Y$, and $\lvert T_Y\rvert^2 \ll (\mathcal{N}/Y)Q_Y^{\mathrm{all}}$.
2	(4.8)–(4.9), p. 16; (4.24)–(4.25), p. 21	Set $t'=t/\mathcal{N}$, use $\psi_1(t/(\mathcal{N}Y))=\psi_1(t'/Y)$ and $\mathfrak{M}_{\mathcal{N}}$. The $\mathcal{N}$ branches cancel the Jacobian $1/\mathcal{N}$ in Lemma 3.1.
3	Proposition 4.3, pp. 20–21; Section 4.6, p. 35	Use labelled index-$\mathcal{N}$ blocks and $P_j=g_{\mathcal{N}}(P)z_j$ with $g_{\mathcal{N}}=gE_{\mathcal{N}}U_0$. At endpoints use $f(Dbm,s)=\overline{\alpha_m}\beta_s$, so $\overline{f(P)}f(Q)=\alpha_m\overline{\alpha_r}\beta_n\overline{\beta_s}$.
4	Continuous extension, p. 22	Keep the entry $\mathcal{N}/u$ exactly in (3.14); $\tau_{\mathcal{N}}(z)=z_1+\beta z_2$ replaces z_1-2rz_2.
5	Path bound, p. 23	Lemma 3.5 directly gives $\lvert z_2\rvert\le 64d_0YMN/\mathcal{N}$, $\lvert z_1\rvert\le 16+\lvert z_2\rvert$. It implies the needed $\lvert z\rvert\ll Rd_0Y$ and coefficient mass $\exp(O_K(L^{0.73}))$.
6	(4.30), (4.32), pp. 23–24	The second boundary is $E_2=t_0\tau_{\mathcal{N}}+z_2\mathcal{N}/(d_0YMNs)$. The extra term is $O(1)$, as is its s derivative; $\partial_{t_0}E_2=\tau_{\mathcal{N}}\in[1/16,16]$ and $\partial_\beta E_2=t_0z_2=\exp(O(L^{0.21}))$. The neighborhood volume is $x^{-c\delta/8+o(1)}$.
7	(4.31), p. 24	$\partial_\beta(\tau_{\mathcal{N}}(w)/\tau_{\mathcal{N}}(z))=1/\tau_{\mathcal{N}}(z)^2$ for $\det(z,w)=1$, replacing $-2/\tau(z)^2$.
8	Boundary cubes, p. 24	Use (s,t_0,β), the same transverse derivatives and $x^{o(1)}$ choices. Sum the interval-uniform errors of Lemma 4.1 over cubes before the moment limit.
9	(4.33)–(4.34), pp. 24–25	Use the primitive relative box of radius $2C_{\mathrm{pos}}Rd_0Y$ and the same τ window. The multiplier is $\chi_{\mathcal{N}}=\mathbf{1}_{g_{\mathcal{N}}z\in\Omega}$, of norm at most 1. Endpoint unit support replaces LWW's z_1-odd test; that test is not implied for the normalized coordinates used here.
10	(4.40)–(4.41), pp. 27–28	$z'=j_1w_z+hz$ and $\tau_{\mathcal{N}}(z')/\tau_{\mathcal{N}}(z)=h+j_1\tau_{\mathcal{N}}(w_z)/\tau_{\mathcal{N}}(z)$. The allowed ratio strip leaves $O(1)$ integers h per j_1.
11	Slope-cell identity, p. 28	$z_2'/\tau_{\mathcal{N}}(z')-z_2/\tau_{\mathcal{N}}(z)=\det(z,z')/(\tau_{\mathcal{N}}(z)\tau_{\mathcal{N}}(z'))$. An edge has reduced determinant $O(d_0Y)$; cells of width d_0Y have bounded interaction overlap. A Goldbach block meets $O(1)$ such cells.
12	(4.42)–(4.46), pp. 28–30	Use the shared-prime lattice, Minkowski box $A'B'\asymp Y$ and the same signed kernel $Y^{-1}\psi_1(\det(k,\ell)/Y)\int_{\mathbb{T}\setminus\mathfrak{M}_1}e(\theta\det(k,\ell))\,d\theta$. Its $\alpha=D\tau_{\mathcal{N}}(w_*)/\tau_{\mathcal{N}}(z_*)$ is $D\beta_{Q_*}$ modulo 1. The enclosing region has $\lvert\ell\rvert\ll Y$ and bounded τ ratios.
13	(4.45), p. 29	$\chi(B_Dk)$ is a bounded endpoint multiplier. Sharp or nonrectangular position restrictions preserve its contraction property.
14	(4.47)–(4.48), p. 30	$j_1=P_{\mathrm{full}}-Dp_*ZT_f=\det(z,z')=D(b-a)$; $\tau_{\mathcal{N}}(z')/\tau_{\mathcal{N}}(z)\in[1/16,16]$. The second good-state test is applied to the intrinsic ratio at the source $g_{\mathcal{N}}(P)z$.

Row	LWW location	Replacement or retained input
15	(4.49)–(4.50), p. 31	The reduced integer box has the same nonzero linear congruence modulo a list prime. List primes are coprime to $\mathcal{N}$ and to the chosen shared D.
16	Endpoint primitivity, p. 35	A common prime of m, s is $> Z$ and coprime to $\mathcal{N}$. It divides $b-a$, while $0 < \lvert b-a\rvert < 4Y < Z$ by the retained cross-edge exclusion. The cofactor sign is irrelevant.
17	Section 4.6 partition, p. 36	Use labelled blocks with $O(d_0Y)$ lattice-primitive positions in the application of Lemma 3.4, with no interaction between blocks. The disjoint endpoint union is restricted to units as in (3.7).
18	(4.63)–(4.68), pp. 36–37	Use the positive principal submatrix on $\mathcal{C}_{\mathcal{N}} \cap \Lambda_\varrho$ in the enlarged ambient space. Spectral Hölder precedes root restriction; $\lambda_{\max} \le d_\varrho$ and $(d_0Y)^{1/(2R)} = O(1)$. Only the restricted-root trace is summed.
19	(4.70), p. 37	Exact-edge $O(1)$ and comparison-edge $O(Y)$ counts are in t'. The bad-state ratio measure is unchanged.
20	(4.74)–(4.76), p. 38; (4.6), p. 16	For fixed $k_0 = mn$, $\ell_0 = rs$ is in the class $ba^{-1}k_0 \bmod \mathcal{N}$ in an interval of length $O(\mathcal{N})$, giving $O(1)$ choices. Raw absolute divisor-square counts use the translated cofactor interval or $\lvert h\rvert \ll \mathcal{N}/Y$, at a cost $\mathcal{N}/X \ll L^B$.
21	Appendix A, swapped case, p. 50	Retain $\mathcal{N}/(uv)$, with $\mathcal{N}/(uv) \ll L^B/(d_0Y)$. Use both $k = \lfloor \beta + \mathcal{N}/(uv)\rfloor \in \{0,1\}$ branches, the separate matrix residues and the cell envelopes of (4.14)–(4.15).
22	Lemma A.1 and (A.5), pp. 47–51	Use Lemma 4.1, the determinant-$\mathcal{N}$ matrix coset, $c_{\mathcal{N}} = \zeta(2)^{-1}\prod_{p\mid\mathcal{N}}(p-1)/(p+1)$ and line masses $1/(p+1)$. Its factor at 2 is already included. Summing matrix errors costs $S^3 = x^{o(1)}$ and decreases the final fixed line-error exponent.
23	Arc measure, p. 16	Haar measure under $\theta \mapsto \mathcal{N}\theta$ gives $\lvert\mathfrak{M}_{\mathcal{N}}\rvert = \lvert\mathfrak{M}_1\rvert \ll L^{3A_0}/Y$.

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Use of AI

The proofs and the text of this paper were produced with AI systems (GPT-6.1 Sol, Claude Fable 5.1 and Claude Opus 5.5) under the author's direction. The proofs were checked by independent reviews with AI models, and the algebraic identities on finite instances by the programs in the ancillary files. The author takes responsibility for the content.